

Factor-Based Neural Network Forecasting of the German Yield Curve



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June 19, 2026

Abstract

This thesis develops a factor-based framework for forecasting the German zero-coupon yield curve. Using daily data from 1999 to 2026, we compare Principal Component Analysis on yield changes and the Nelson-Siegel model as dimensionality reduction approaches, finding that PCA with four components provides superior and more stable reconstruction across all maturities and yield curve regimes. Rolling PCA loadings are treated as a locally stable coordinate system, ensuring that lagged factor inputs and forecast targets share a common basis across the full lag horizon.

We estimate a vector autoregression over the four principal component changes as the linear benchmark, establishing that the components are dynamically interdependent despite their cross-sectional orthogonality. An LSTM network is then evaluated across five variants with increasing information sets. We find that feeding the LSTM the coordinate-consistent factor representation, rather than raw factor differences, is essential for performance, and that the richest variant incorporating swaption implied volatility achieves the only positive out-of-sample R^2 for the fourth principal component across all specifications, though it does not surpass the VAR overall.

Reconstructing the yield curve directly from these factor forecasts overstates the magnitude of yield changes and underperforms the random walk at the one-month horizon. We address this with a two-parameter scaling procedure that separately calibrates the momentum and predicted-change components of the reconstruction, with the scaling coefficients estimated on a rolling out-of-sample basis. For the VAR, both coefficients are positive in roughly 90% of rolling windows, and the resulting scaled reconstruction beats the random walk on RMSE across the short and medium end of the curve (1Y–15Y), with directional accuracy of approximately 54–60% per tenor. By contrast, the LSTM’s scaling coefficients are negative throughout, so its factor-level gains do not survive reconstruction. The stability of the scaling coefficients over time further provides a practical, interpretable signal for when the reconstruction can be relied upon.

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1 Introduction

The yield curve, describing the relationship between interest rates and time to maturity for government bonds, plays a central role in monetary policy and fixed-income markets. For the European Central Bank, with whom this research is conducted, the German zero-coupon yield curve serves as an internal fair value model: constructed daily from all outstanding German government bonds, it provides the benchmark against which individual securities are assessed as cheap or expensive relative to the market. Anticipating which parts of the term structure are likely to shift over the coming month gives direct insight into where mispricing may emerge and supports forward-looking intelligence on interest rate developments.

This thesis develops a factor-based forecasting framework for that curve. We first establish that PCA on yield changes dominates Nelson-Siegel in reconstruction accuracy and adopt it as the basis for forecasting. We then formalise the treatment of rolling PCA loadings as a stable coordinate system, ensuring that all model inputs are expressed in a common basis. We estimate a vector autoregression over the resulting factor series as the linear benchmark, and evaluate whether an LSTM network can capture nonlinear dynamics beyond it. We then map the resulting factor forecasts back to yield curve levels through a two-parameter scaling procedure, estimated on a rolling out-of-sample basis, and assess whether the reconstructed curve improves on a random walk benchmark. Finally, we investigate whether swaption implied volatility carries incremental predictive information for factor movements.

Dimensionality reduction is the natural starting point for yield curve forecasting. Litterman and Scheinkman (1991) establish empirically that three principal components account for nearly all yield curve variation, corresponding to level, slope, and curvature. Nelson and Siegel (1987) propose a parametric alternative that represents the same structure with three factors governed by a single decay parameter λ . Both approaches motivate the same two-stage procedure used by Diebold and Li (2006): extract a small number of factors, then forecast them as a time series. The key difference is that PCA imposes no assumption on the shape of factor loadings and produces orthogonal factors by construction, which avoids the estimation instability documented for four-factor parametric extensions Gilli et al. (2010). We evaluate both methods on German zero-coupon yield data and find PCA on yield changes to be the superior representation.

We also examine the stability of the rolling PCA loadings over time. The VAR, which ultimately proves to be the stronger model, is built directly on the lagged factor differences $\Delta f_{k,t-\ell}$, each of which is implicitly expressed in the loading matrix prevailing at $t - \ell$ rather than a common basis. We demonstrate empirically that the loading matrix $\mathbf{L}_t$ changes negligibly over a one-month horizon, which means that, in the other direction, these lagged factors are also approximately comparable over time even without an explicit re-projection. The same stability result allows lagged observations to instead be projected onto a single $\mathbf{L}_t$, ensuring that inputs and forecast targets share a common basis. This coordinate-consistent representation matters for the LSTM in particular: replacing raw factor differences with it improves directional accuracy and out-of-sample R^2 for two of the four components, suggesting that input representation, rather than architecture alone, may be a meaningful driver of LSTM performance.

To exploit the autocorrelation structure in the factor series, we estimate a vector autoregression following Diebold and Li (2006). A non-obvious finding is that the four principal components,

while orthogonal by construction, are dynamically interdependent: lagged values of the level factor are significant predictors in the equations for all other components, making the joint VAR the appropriate specification over univariate alternatives. The VAR proves to be a strong benchmark throughout this thesis and ultimately forms the basis of the final reconstruction.

To capture potential nonlinear dynamics beyond the VAR, we employ a Long Short-Term Memory network Hochreiter and Schmidhuber (1997); Gers (1999). Using the coordinate consistent representation, the LSTM outperforms the VAR on directional accuracy and out-of-sample R^2 for the second and fourth principal components, indicating that nonlinear dynamics are present and detectable for these factors. At the yield curve level, however, the LSTM's predictions do not translate into a stable or economically coherent scaling, and the model does not improve on the VAR once reconstructed. Whether this reflects a genuine limit on the predictability of these dynamics or simply that the specific architecture employed here is not optimally configured remains an open question. A further advantage of the LSTM framework, distinct from its predictive performance, is its capacity to incorporate external variables alongside the factor history, which the VAR cannot accommodate without a substantial increase in dimensionality.

Finally, motivated by Verner et al. (2021), who examine external market variables in fixed-income neural network forecasting, we test whether the SMOVE swaption implied volatility index, a yield-curve-weighted measure of normalised implied volatility on 3-month swaptions across the 2-, 5-, 10-, and 30-year swap maturities, provides incremental predictive information for factor movements.

The central result of this thesis concerns the reconstruction of yield curve forecasts from the factor predictions. We show that a naive reconstruction overstates the magnitude of yield changes, and address this through a two-parameter scaling procedure that separately calibrates the momentum and predicted-change components of the forecast. Estimated on a rolling out-of-sample basis, this scaled VAR reconstruction beats the random walk on RMSE across most of the curve and achieves directional accuracy of approximately 54–60% per tenor over the recent out-of-sample period, while the stability of the estimated scaling parameters provides a practical signal for when the model can be relied upon.

The remainder of this thesis is organised as follows. Section 2 reviews the relevant literature. Section 3 describes the data. Section 4 compares the dimensionality reduction approaches and validates the loading stability assumption. Section 5 presents the VAR forecasting results. Section 6 develops the LSTM framework and evaluates its out-of-sample performance against the VAR. Section 7 reconstructs yield curve forecasts from the component-level predictions via the alpha-scaling procedure and evaluates them against a random walk benchmark. Section 8 concludes.

2 Literature Review

The literature on yield curve modeling and forecasting has evolved considerably over the past decades, progressing from parametric curve-fitting methods to factor-based forecasting frameworks, and more recently incorporating machine learning techniques. This review traces this evolution, beginning with foundational work on dimensionality reduction, proceeding through the development of dynamic factor forecasting models, and concluding with recent applications of neural networks to yield curve analysis. This review is structured to highlight how each contribution addresses specific aspects of the yield curve forecasting problem.

2.1 Dimensionality Reduction

Forecasting the yield curve presents an inherent high-dimensionality challenge: yields across many maturities must be predicted simultaneously, yet modelling each maturity independently ignores the strong co-movement that characterises term structure dynamics. A small number of common factors explains the vast majority of yield variation Litterman and Scheinkman (1991), making dimensionality reduction a natural first step that transforms the forecasting problem from predicting many correlated yields to predicting a handful of interpretable factors.

2.1.1 Parametric Dimensionality Reduction: The Nelson-Siegel Model and Its Extensions

The challenge of modeling yield curves across numerous maturities was first addressed systematically by Nelson and Siegel (1987), who proposed a parsimonious functional form capable of generating the diverse shapes observed in practice. The Nelson-Siegel model represents the zero-coupon yield curve at time t for maturity τ as:

$$y_t(\tau) = \beta_{1,t} + \beta_{2,t} \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} \right) + \beta_{3,t} \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right) \quad (1)$$

where $\beta_{1,t}$, $\beta_{2,t}$, and $\beta_{3,t}$ are time-varying parameters representing level, slope, and curvature respectively, and λ is a fixed decay parameter. By reducing an N -dimensional yield curve to just three parameters, Nelson and Siegel (1987) demonstrated that complex yield curve shapes could be parsimoniously represented while maintaining economic interpretability.

Svensson (1994) extended this framework by introducing a fourth factor with an additional curvature term, increasing flexibility in fitting irregular curve shapes. This Nelson-Siegel-Svensson (NSS) model has become standard practice among central banks BIS (2005). However, the two curvature factors are prone to confounding effects, where the parameters interact and can take on extreme values to offset one another, producing unstable estimates Gilli et al. (2010). This instability motivates the use of the three-factor Nelson-Siegel specification in this thesis rather than its four-factor extension, as discussed further in Section 4.

2.1.2 Statistical Dimensionality Reduction: Principal Component Analysis

Litterman and Scheinkman (1991) took a different approach by applying principal component analysis (PCA) to U.S. Treasury yields, letting the data reveal the dominant factors driving yield curve variation rather than imposing a parametric structure. The PCA approach offers two advantages over parametric methods. First, it requires no assumption about the

functional form of factor loadings. Second, it provides a natural measure of how much variance each factor explains, allowing researchers to determine the appropriate number of factors empirically. Crucially, unlike the NSS model, PCA factors are orthogonal by construction, which prevents the parameter explosion documented in the parametric setting Gilli et al. (2010). This makes it natural to retain a fourth principal component without the instability concerns associated with the NSS fourth factor, a choice we revisit and justify empirically in Section 4.

2.2 Dynamic Factor Forecasting: The Diebold-Li Framework

While Nelson and Siegel (1987) and Litterman and Scheinkman (1991) established that yield curves can be effectively represented by three factors, Diebold and Li (2006) made the crucial step of creating a forecasting framework by treating these factors as time series to be predicted. Diebold and Li (2006) combined the Nelson and Siegel (1987) factor structure with vector autoregression (VAR) models to forecast the entire yield curve. Their two-stage approach works as follows. First, at each point in time, they estimate the three Nelson and Siegel (1987) factors $\beta_{1,t}$, $\beta_{2,t}$, and $\beta_{3,t}$ by fitting equation (1) to observed yields. Second, they model the dynamics of these factors using a VAR(1) specification:

$$\begin{bmatrix} \beta_{1,t} \\ \beta_{2,t} \\ \beta_{3,t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} & \phi_{13} \\ \phi_{21} & \phi_{22} & \phi_{23} \\ \phi_{31} & \phi_{32} & \phi_{33} \end{bmatrix} \begin{bmatrix} \beta_{1,t-1} \\ \beta_{2,t-1} \\ \beta_{3,t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \varepsilon_{3,t} \end{bmatrix} \quad (2)$$

To generate forecasts h periods ahead, they iterate the VAR forward and then reconstruct the yield curve at each horizon using the Nelson and Siegel (1987) formula 1 with the forecasted factor values. This approach transforms a high-dimensional forecasting problem (predicting yields at many maturities) into a low-dimensional one (predicting three factors).

The key empirical finding of Diebold and Li (2006) is that this factor-based VAR approach significantly outperforms a simple random walk benchmark in out-of-sample forecasts, particularly at medium to long horizons. The random walk model, had proven difficult to beat in earlier research, making this result notable. The success of the Diebold and Li (2006) framework stems from exploiting the cross-sectional restrictions implied by the factor structure: rather than forecasting each maturity independently, the model recognizes that yields across maturities move together in predictable ways captured by the three factors. This framework has become the benchmark against which subsequent yield curve forecasting methods are evaluated.

However, the Diebold and Li (2006) approach has an important limitation: the VAR specification assumes that the relationship between past and future factor values is linear. While this assumption simplifies estimation and interpretation, it may be restrictive if yield curve dynamics exhibit regime changes, threshold effects, or other nonlinearities. Moreover, the model relies exclusively on the historical behavior of the factors themselves, ignoring potentially valuable information from other market variables that might help predict future movements.

2.3 Neural Networks for Yield Curve and Financial Time Series Forecasting

The application of neural networks to yield curve and broader financial time series forecasting has grown rapidly, and the resulting evidence is mixed: some studies report clear out-of-sample gains over linear or memory-free benchmarks, while others find that added architectural complexity contributes little once a reasonable baseline is in place. This section reviews the LSTM architecture itself, a recent extension of the Diebold-Li framework that incorporates machine learning directly, what the literature says about architectural choice versus input representation, and the specific prior application to yield curve forecasting that motivates the design adopted in this thesis.

2.3.1 Long Short-Term Memory

Hochreiter and Schmidhuber (1997) introduced the Long Short-Term Memory (LSTM) network to address the vanishing gradient problem that prevented standard recurrent neural networks from learning dependencies across long time intervals. The key innovation is the memory cell, which maintains its state across timesteps through a constant error carousel, with multiplicative input and output gates controlling access to the stored information. The original architecture was subsequently extended by Gers (1999), who identified a weakness in processing continuous input streams: without a mechanism to reset cell state, the memory can grow indefinitely and cause network saturation. Their remedy is the forget gate, completing the three-gate architecture that is standard in modern implementations including TensorFlow and employed in this thesis. A useful way to view this gating structure is as a form of dynamic information filtering: at every timestep, the forget and input gates decide which information from the factor history remains relevant and which can be discarded, continuously and as part of training, rather than through any externally imposed rule. This is conceptually related to, though mechanically quite different from, the static partitioning of Bie et al. (2024) discussed next, where a single, fixed split of the sample determines once which past information is treated as relevant. With the LSTM architecture itself fixed, the remaining open question is whether any further choice, of variant, depth, or hyperparameter, materially affects performance, or whether the more consequential choice lies elsewhere.

2.3.2 Machine Learning Within the Diebold-Li Framework: Bie et al. (2024)

A direct link between the dynamic factor approach of Section 5 and the machine learning literature is provided by Bie et al. (2024), who extend the dynamic Nelson-Siegel framework of Diebold and Li (2006) with macro-instrumented regime switching. Their factor dynamics remain a VAR(1), as in Diebold and Li (2006); the machine learning component is a decision-tree splitting procedure that partitions the sample, once and ex ante, into regimes defined by thresholds on observed macroeconomic variables, principally the federal funds rate, with split points chosen to maximise the model’s marginal likelihood, and a separate VAR estimated within each regime. Applied to U.S. Treasury yields from 1971 to 2022, they find that macroeconomic variables carry predictive power for the yield curve specifically in the regime with a high federal funds rate, but little or no such power in the other two regimes, refining the broader notion of yield-curve “macro-spanning” into something regime-dependent rather than universally true or false.

This regime-dependence is directly relevant to a pattern that recurs later in this thesis: if the predictive content of an exogenous variable is itself conditional on regime, an evaluation

window spanning only some regimes would show benefits concentrated in particular periods or components, as observed for the SMOVE swaption volatility feature in Section 6.4, without this implying the variable is uninformative in general. More broadly, both this paper and the present thesis treat model reliability as a quantity to be estimated and monitored rather than assumed, theirs through a fixed regime classification and this thesis through the rolling scaling coefficient $\hat{\alpha}(t)$ introduced in Section 7.3. This thesis does not itself adopt tree-based regime switching; the connection drawn here is to the shared principle of monitoring model reliability rather than to a shared mechanism.

2.3.3 Evidence on What Drives LSTM Performance

Returning to the open question of Section 2.3.1, two strands of evidence suggest that architectural variation within the LSTM family is not where the literature locates its gains. Greff et al. (2015) conduct the largest study of its kind on LSTM networks, evaluating eight architectural modifications across three sequence learning tasks in 5,400 experimental runs, and find that none of the eight variants improves significantly on the standard LSTM, with only the forget gate and output activation function identified as critical components, while training hyperparameters such as learning rate and network size behave as approximately independent of one another. Ayitey Junior et al. (2022) compare an attention-augmented LSTM (ALFA) against several recurrent baselines, including a stacked LSTM, GRU, and bidirectional LSTM, on forex rate prediction, and find that while the plain recurrent baselines are themselves competitive, the consistent top-three performance of ALFA is attributable specifically to its attention mechanism, which reweights the network’s output features toward the most relevant ones, rather than to depth or recurrence alone. Read together, these results suggest that complexity for its own sake, whether through gating variants or stacking, is not reliably rewarded, but complexity targeted at improving what information the network has access to can be. This is the seam this thesis enters through: rather than searching across LSTM variants for which the evidence above finds limited benefit, Section 6.2 fixes the architecture and instead varies the input representation fed to it, treating the resulting performance differences as the object of study.

A related question is whether LSTM gains, where they occur, are understood or merely observed. Fischer and Krauss (2018) apply LSTMs to predict daily directional stock movements for S&P 500 constituents over 1992–2015, find that the LSTM outperforms a random forest, a standard feedforward network, and a logistic regression on both accuracy and realised returns, and then decompose the source of this edge: the stocks the LSTM selects share identifiable characteristics, namely high volatility and short-term reversal, and a simplified rules-based strategy built on these characteristics alone recovers a substantial share of the LSTM’s returns, leaving only a residual attributable to genuinely higher-order dynamics. Nunes et al. (2026) pursue a complementary, mechanistic version of the same question for bond yields specifically: their LSTM-LagLasso framework regresses a trained network’s internal gating signals on candidate exogenous variables at multiple lags, and finds that individual hidden units specialise, tracking distinct non-stationarities in the yield series and responding to different variables and lag structures, while also achieving better accuracy than a multilayer perceptron benchmark. Both results suggest that an LSTM’s apparent edge can often be partially attributed to identifiable, even linear, structure once the model is opened up, with only a residual genuinely requiring the network’s nonlinear capacity. This motivates how this thesis interprets its own results in Section 6.4: where a richer input representation improves performance for only some principal components rather than all four, a gate-level

inspection along the lines of Nunes et al. (2026) would be needed to confirm the mechanism, and is noted there as a direction for future work rather than attempted directly.

2.3.4 LSTM-Based Yield Curve Forecasting: Suimon et al. (2020)

Suimon et al. (2020) provide the most directly comparable prior application to this thesis. Their primary contribution is a three-factor autoencoder model for the Japanese government bond yield curve, in which the autoencoder’s intermediate-layer parameters are shown to recover factors interpretable as level, slope, and curvature without imposing this structure a priori. In addition, Suimon et al. (2020) implement an LSTM that forecasts the yield curve one month ahead directly from historical interest rate data, taking as input the full yield curve across six maturities at four consecutive weekly timesteps, and find that this LSTM strategy outperforms both the autoencoder-based valuation strategy and a trend-following benchmark in cumulative capital gains, particularly for the 10- and 20-year maturities. Notably, they report that a VAR model performs comparably to the LSTM on prediction accuracy, suggesting that much of the exploitable signal in their data is already linear, with the LSTM contributing a further but modest improvement, and they note a genuine trade-off in their results: the autoencoder is more interpretable while the LSTM is more accurate, a trade-off they leave for future work to reconcile.

This thesis adopts the same core idea, an LSTM operating on a short sequence of historical yield curve information to produce a one-month-ahead forecast, but departs from Suimon et al. (2020) in two respects. First, rather than feeding the LSTM raw yields directly, this thesis forecasts PCA factor changes under the coordinate-consistent representation of Section 4.5, so that the cross-sectional structure established in Sections 4 and 5 is preserved rather than re-learned by the network. Second, this thesis stacks two LSTM layers rather than using the single-layer specification of Suimon et al. (2020), a choice made because the evidence reviewed above gives no indication that stacking degrades performance relative to a single layer: Greff et al. (2015) find architectural variants within the LSTM family to be largely interchangeable, while Ayitey Junior et al. (2022) find that, where additional capacity did improve on plain recurrent baselines including a stacked LSTM, the improvement was attributable to attention rather than to depth itself. The full justification for the resulting two-layer configuration is given in Section 6.3.

3 Data

3.1 Yield Curve Data

The zero-coupon yield curve data for Germany are obtained from an internal ECB database. The curve is constructed by collecting all outstanding German government bonds at each point in time from the IBoxx bond database, a product of LSEG (formerly Markit), computing zero-coupon yields from observed bond prices, and fitting a locally weighted scatterplot smoothing (LOWESS) spline Cleveland (1979) across maturities to obtain a continuous term structure. This curve constitutes the ECB's fair value model of the German term structure.

The dataset spans 4 January 1999 to 20 April 2026, comprising 6,986 daily observations after excluding weekends. Yields are available at 15 maturity points: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 15, 20, 25, and 30 years. No missing observations are present in the yield curve series.

Table 1 presents summary statistics across all maturities. Where Figure 1 illustrates the full evolution of the yield curve over the sample period.

Table 1: Summary statistics of German zero-coupon yields of ECB fair value model, 1999–2026. $N = 6,986$ daily observations.

Maturity	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
1Y	0.015	0.018	−0.009	−0.002	0.015	0.030	0.051
2Y	0.016	0.018	−0.010	−0.002	0.017	0.030	0.052
3Y	0.017	0.018	−0.010	−0.001	0.019	0.030	0.052
4Y	0.018	0.018	−0.010	−0.001	0.021	0.033	0.052
5Y	0.019	0.018	−0.010	0.000	0.022	0.035	0.053
6Y	0.020	0.018	−0.010	0.002	0.023	0.036	0.053
7Y	0.021	0.018	−0.010	0.003	0.023	0.037	0.055
8Y	0.023	0.018	−0.009	0.004	0.024	0.039	0.055
9Y	0.023	0.018	−0.009	0.005	0.025	0.039	0.056
10Y	0.024	0.018	−0.009	0.007	0.026	0.040	0.056
12Y	0.026	0.017	−0.008	0.008	0.027	0.041	0.057
15Y	0.027	0.017	−0.008	0.010	0.028	0.042	0.058
20Y	0.029	0.018	−0.007	0.012	0.030	0.045	0.059
25Y	0.031	0.019	−0.006	0.013	0.030	0.047	0.068
30Y	0.030	0.018	−0.005	0.014	0.030	0.045	0.063

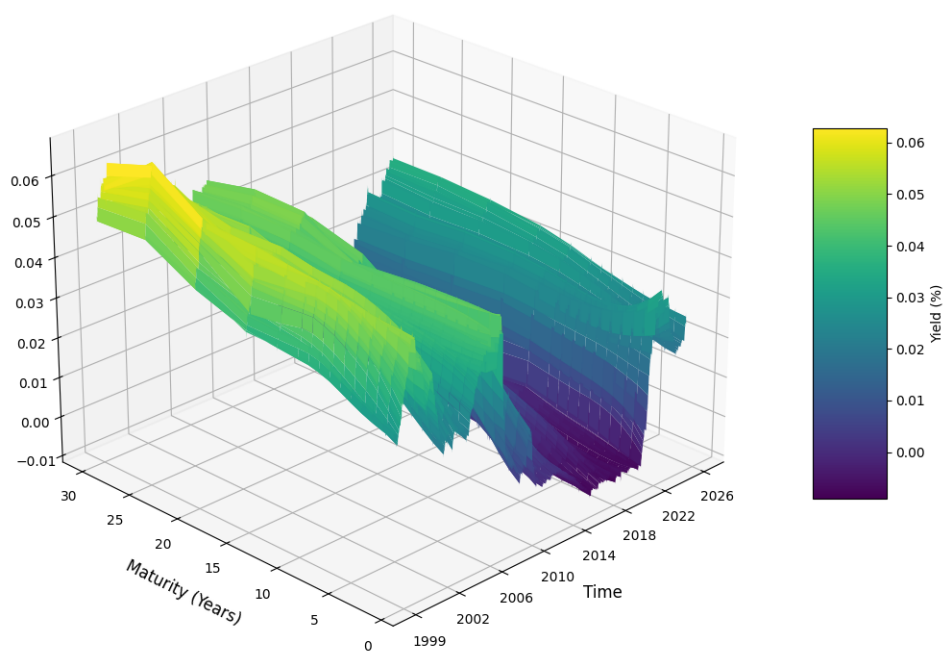


Figure 1: Evolution of the German zero-coupon yield curve, 1999–2026.

3.2 Swaption Implied Volatility Data

The implied volatility series is the Merrill Lynch Swaption Option Volatility Estimate (SMOVE) for EUR, sourced from Bloomberg Finance L.P. and obtained via an internal ECB database. SMOVE is a yield curve weighted index of normalised implied volatility on 3-month swaptions, computed as the weighted average of implied volatilities on 2-year, 5-year, 10-year, and 30-year maturity swaps. The volatility is expressed in basis points per annum.

The series spans the same sample period as the yield curve data. For 96 dates where observations are missing, values are recovered by linear interpolation between the nearest available observations before and after the gap:

$$IV_t = IV_{t-k} + \frac{t - (t-k)}{(t+j) - (t-k)} (IV_{t+j} - IV_{t-k}) \quad (3)$$

where k and j denote the number of days to the nearest observed values before and after t . All gaps are short, consisting of isolated missing dates, so interpolation introduces negligible distortion.

Table 2 presents summary statistics, where Figure 2 shows the full time series.

Table 2: Summary statistics of $3M \times 10Y$ ATM normal swaption implied volatility, 1999–2026. Values in basis points. $N = 6,986$.

	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
IV (bp)	66.7	25.1	26.3	49.6	62.5	79.5	180.3

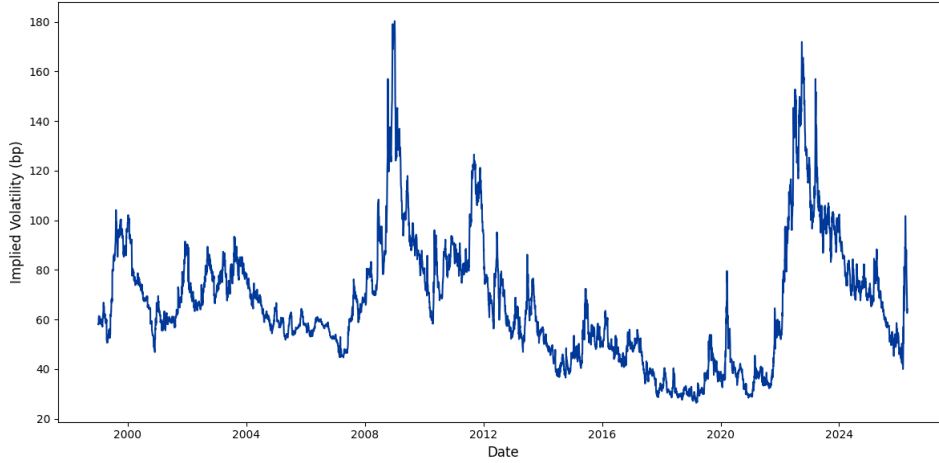


Figure 2: $3M \times 10Y$ ATM normal swaption implied volatility, 1999–2026. Values in basis points.

4 Yield Curve Fitting

4.1 Motivation

We evaluate two approaches: Principal Component Analysis on yield changes and the Nelson-Siegel model. PCA is applied to yield *changes* rather than levels, as stationarity of the input series is required for the sample covariance matrix to converge. Throughout, the forecast horizon is $h = 21$ trading days, which approximates a forecast of one-month-ahead.

Motivated by the four-factor extension of Svensson (1994), we test both three and four principal components. We do not consider the four-beta Nelson-Siegel- Svensson model, as Gilli et al. (2010) document severe optimisation instability in the NSS parameterisation: the additional decay parameter renders the loading matrix ill-conditioned, making reliable rolling estimation without additional constraints infeasible. PCA's orthogonality constraint avoids this issue, allowing a fourth factor to be retained without instability. The following subsections develop the mathematical framework for each method and evaluate their empirical fit on German zero-coupon yield data.

4.2 Nelson-Siegel

The Nelson-Siegel model Nelson and Siegel (1987) represents the yield curve parametrically as:

$$y(M) = \beta_0 + \beta_1 \frac{1 - e^{-\lambda M}}{\lambda M} + \beta_2 \left(\frac{1 - e^{-\lambda M}}{\lambda M} - e^{-\lambda M} \right) \quad (4)$$

where M denotes maturity and $\lambda > 0$ controls the decay rate of the loadings. In matrix form, for $m = 15$ maturities:

$$\mathbf{y}_t = \mathbf{L}(\lambda_t) \boldsymbol{\beta}_t \quad (5)$$

where $\mathbf{y}_t \in \mathbb{R}^m$, $\mathbf{L}(\lambda_t) \in \mathbb{R}^{m \times 3}$ contains the maturity-dependent loadings fully determined by λ_t , and $\boldsymbol{\beta}_t = [\beta_0, \beta_1, \beta_2]^T$ are the factor weights estimated by OLS given λ_t . The optimal λ_t is estimated by minimising reconstruction error over a rolling window of $\tau = 756$ trading days.

Forecasting $h = 21$ days ahead requires that the loading matrix remains approximately stable over the forecast horizon:

$$\lambda_{t+h} \approx \lambda_t \quad (6)$$

Since $\mathbf{L}(\lambda_t)$ is fully determined by λ_t , this is equivalent to $\mathbf{L}(\lambda_{t+h}) \approx \mathbf{L}(\lambda_t)$. Given a stable λ_t , future factor weights $\boldsymbol{\beta}_{t+h}$ are refitted by OLS at $t+h$ and the yield curve is reconstructed as:

$$\hat{\mathbf{y}}_{t+h} = \mathbf{L}(\lambda_t) \boldsymbol{\beta}_{t+h}(\lambda_t) \quad (7)$$

Assumption (6) is tested empirically in Section 4.6.

4.3 Principal Component Analysis on Yield Changes

PCA is applied to yield changes rather than levels, as stationarity of the input series is required for the sample covariance matrix to converge properly. The covariance matrix is estimated from daily yield changes $\delta \mathbf{y}_t = \mathbf{y}_t - \mathbf{y}_{t-1} \in \mathbb{R}^m$ over a rolling window of $\tau = 756$ trading days:

$$\hat{\boldsymbol{\Sigma}}_t^\delta = \frac{1}{\tau - 1} \sum_{s=t-\tau}^{t-1} (\delta \mathbf{y}_s - \overline{\delta \mathbf{y}}_t)(\delta \mathbf{y}_s - \overline{\delta \mathbf{y}}_t)^T \quad (8)$$

Rather than fitting on overlapping 21-day changes, where consecutive observations share 20 out of 21 days and introduce artificial serial correlation, we fit on daily changes and scale under the i.i.d. assumption:

$$\hat{\Sigma}_t^\Delta = 21 \cdot \hat{\Sigma}_t^\delta \quad (9)$$

yielding 756 approximately independent observations per estimate. The eigendecomposition of $\hat{\Sigma}_t^\Delta$ gives:

$$\hat{\Sigma}_t^\Delta = \mathbf{V}_t \mathbf{\Lambda}_t \mathbf{V}_t^T \quad (10)$$

where $\mathbf{\Lambda}_t = \text{diag}(\nu_{1,t}, \dots, \nu_{m,t})$ contains eigenvalues in descending order. Since multiplying by a scalar does not alter eigenvectors, $\mathbf{V}_t$ is identical whether computed from daily or 21-day changes. The loading matrix retains the k leading eigenvectors:

$$\mathbf{L}_t = [\mathbf{v}_{1,t} \ \mathbf{v}_{2,t} \ \dots \ \mathbf{v}_{k,t}] \in \mathbb{R}^{m \times k} \quad (11)$$

where $\mathbf{v}_{j,t}$ denotes the j -th eigenvector of $\hat{\Sigma}_t^\Delta$ ordered by descending eigenvalue. We consider $k \in \{1, 2, 3\}$ for PCA3 and $k \in \{1, 2, 3, 4\}$ for PCA4, the two specifications are referred to as PCA3 and PCA4 throughout.

Factors are extracted by projecting the 21-day yield change $\Delta \mathbf{y}_t = \mathbf{y}_t - \mathbf{y}_{t-21}$ onto $\mathbf{L}_t$:

$$\mathbf{f}_t = \mathbf{L}_t^T \Delta \mathbf{y}_t \in \mathbb{R}^k \quad (12)$$

and the change is approximated by projecting back:

$$\widetilde{\Delta \mathbf{y}_t} = \mathbf{L}_t \mathbf{f}_t = \mathbf{L}_t \mathbf{L}_t^T \Delta \mathbf{y}_t \quad (13)$$

Since $\mathbf{L}_t$ has orthonormal columns, $\mathbf{L}_t^T \mathbf{L}_t = \mathbf{I}_k$, so $\mathbf{L}_t \mathbf{L}_t^T$ is the orthogonal projection onto the k -dimensional subspace spanned by the retained eigenvectors. The yield curve level is then recovered as:

$$\hat{\mathbf{y}}_{t+h} = \mathbf{y}_t + \mathbf{L}_t \mathbf{f}_{t+h} \quad (14)$$

where $\mathbf{f}_{t+h} = \mathbf{L}_t^T \Delta \mathbf{y}_{t+h}$ are the factors refitted at $t+h$ under the stale loading matrix $\mathbf{L}_t$. Forecasting $h = 21$ days ahead requires that the loading matrix remains approximately stable:

$$\mathbf{L}_{t+h} \approx \mathbf{L}_t \quad (15)$$

This assumption is tested empirically in Section 4.6.

4.4 Autocorrelation Analysis and Lag Selection

For each component $k \in \{1, 2, 3, 4\}$ under the PCA4 model, the target series is the 21-trading-day change in the k -th principal component:

$$\Delta f_{k,t} = f_{k,t+h} - f_{k,t}, \quad h = 21 \quad (16)$$

To identify the dependence structure, we compute the autocorrelation function (ACF) and partial autocorrelation function (PACF). The ACF at lag ℓ is:

$$\hat{\rho}(\ell) = \frac{\sum_{t=\ell+1}^N (\Delta f_{k,t} - \overline{\Delta f_k})(\Delta f_{k,t-\ell} - \overline{\Delta f_k})}{\sum_{t=1}^N (\Delta f_{k,t} - \overline{\Delta f_k})^2} \quad (17)$$

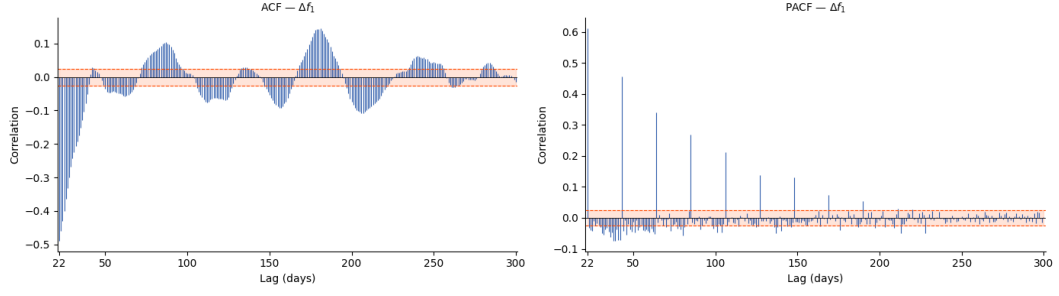
The PACF at lag ℓ isolates the direct effect of lag ℓ after conditioning on all intermediate lags $1, \dots, \ell - 1$, obtained via the Yule-Walker equations Lütkepohl (2005). A sharp PACF cutoff with gradual ACF decay is the identifying signature of an autoregressive process. Significance is assessed against the 95% confidence bound $\pm 1.96/\sqrt{N}$.

Because $\Delta f_{k,t}$ is a cumulative h -period change, consecutive observations share $h - 1 = 20$ overlapping days, mechanically contaminating the ACF and PACF for lags $\ell < h = 21$. The first interpretable lag is therefore $\ell^* = h + 1 = 22$, coinciding with the first element of $\mathcal{L}$ introduced in Section 4.5.

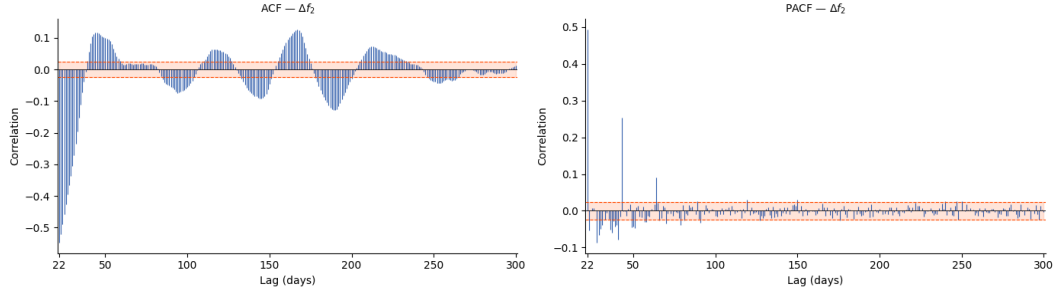
Figure 3 presents the results. The PACF shows a large positive spike at $\ell^* = 22$ for all four components, with values between 0.46 and 0.61, confirming strong direct mean-reversion at the first valid lag. Significant partial autocorrelations persist to lag 211 for Δf_1 , to approximately lag 127 for Δf_2 and Δf_3 , and with intermediate persistence for Δf_4 . The gradual ACF decay without a clean cutoff is consistent with an autoregressive rather than moving average structure. Based on these findings, we define the forecastable lag set:

$$\mathcal{L}^* = \{22, 43, 64, 85, 106, 127, 148, 169, 190, 211\}$$

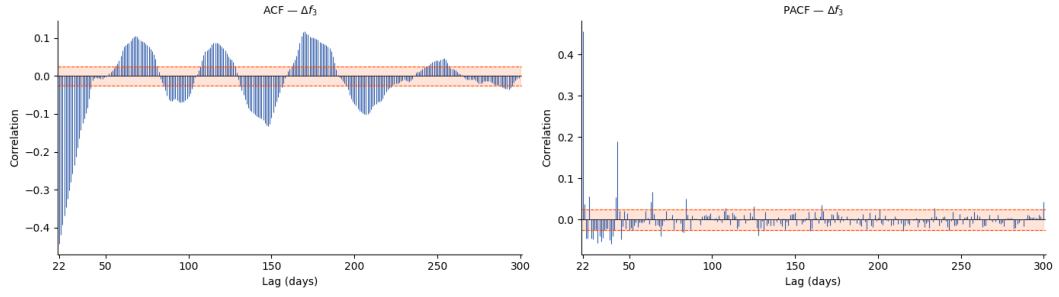
capturing the full persistence horizon of Δf_1 without unnecessarily extending the model. Note that $\mathcal{L}^*$ excludes lag 0 as the current observation cannot serve as a predictor of itself. Full ACF and PACF values at all lags in $\mathcal{L}^*$ are reported in Appendix A3.



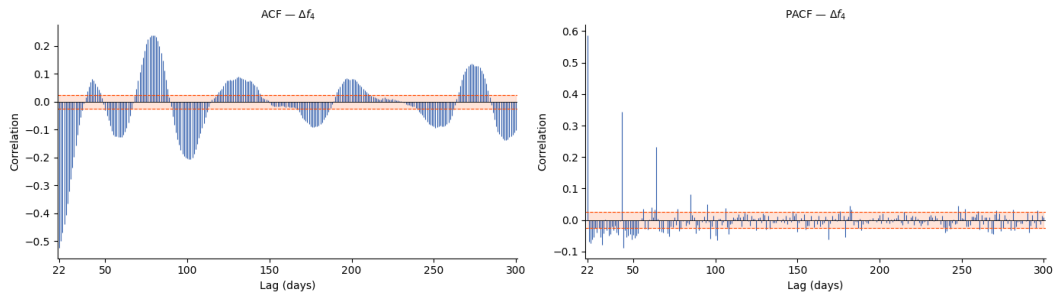
(a) Δf_1



(b) Δf_2



(c) Δf_3



(d) Δf_4

Figure 3: ACF (left) and PACF (right) of 21-day principal component changes $\Delta f_{k,t}$ for $k \in \{1, 2, 3, 4\}$. Dashed lines indicate 95% confidence bounds $\pm 1.96/\sqrt{N}$.

4.5 A Consistent Factor Coordinate System

For the forecasting models developed in subsequent sections, lagged factor values are used as input features. Let $\mathcal{L} = \{0, 22, 43, 64, 85, 106, 127, 148, 169, 190, 211\}$ denote the set of positions in trading days, where lag 0 denotes the current observation and is included for reconstruction and stability assessment only. The non-zero values correspond to $\mathcal{L}^*$, motivated by the autocorrelation structure documented in Section 4.4. For each lag $\ell \in \mathcal{L}$, the 21-day yield change is projected onto the *current* loading matrix $\mathbf{L}_t$:

$$\mathbf{f}_{t-\ell} = \mathbf{L}_t^T \Delta \mathbf{y}_{t-\ell}, \quad \ell \in \mathcal{L} \quad (18)$$

Projecting all lagged observations onto the same $\mathbf{L}_t$ ensures that $\mathbf{f}_{t-\ell}$ across all lags are expressed in a common basis and might not be directly comparable otherwise. The quality of this representation at each lag is measured by:

$$\text{RMSE}_{t-\ell} = \sqrt{\frac{1}{m} \|\Delta \mathbf{y}_{t-\ell} - \mathbf{L}_t \mathbf{L}_t^T \Delta \mathbf{y}_{t-\ell}\|^2}, \quad \ell \in \mathcal{L} \quad (19)$$

This extends the stability assumption (15) beyond the forecast horizon $h = 21$ to the full lag horizon of 211 trading days: reliable factor representations require $\mathbf{L}_t$ to remain a valid basis not only at $t + h$ but also at $t - \ell$ for all $\ell \in \mathcal{L}$. The empirical validity of this assumption across all lag positions and the forecast horizon is tested in Section 4.6. Among the models investigated for the forecasting problem, an LSTM is evaluated in Section 6, where the value of this lag-consistent representation is assessed directly.

4.6 Loading Stability

We test the stability assumptions (15) and (6) empirically across all positions $p \in \mathcal{P}$, where $\mathcal{P} = \mathcal{L} \cup \{+h\}$ covers both the lagged positions and the forecast horizon $h = 21$. For PCA3 and PCA4 the reconstruction RMSE at each position follows (19). The analogous quantity for Nelson-Siegel uses the stale λ_t with factor weights refitted at each position:

$$\text{RMSE}_{t,p}^{\text{NS}} = \sqrt{\frac{1}{m} \|\mathbf{y}_{t+p} - \mathbf{L}(\lambda_t) \boldsymbol{\beta}_{t+p}(\lambda_t)\|^2}, \quad p \in \mathcal{P} \quad (20)$$

Since (20) operates on yield levels while (19) operates on yield changes, the absolute magnitudes across methods are not directly comparable. Both are nonetheless valid stability tests individually: flat profiles across positions indicate that the stale loading structure represents the yield curve consistently over time, including at the forecast horizon $t + h$.

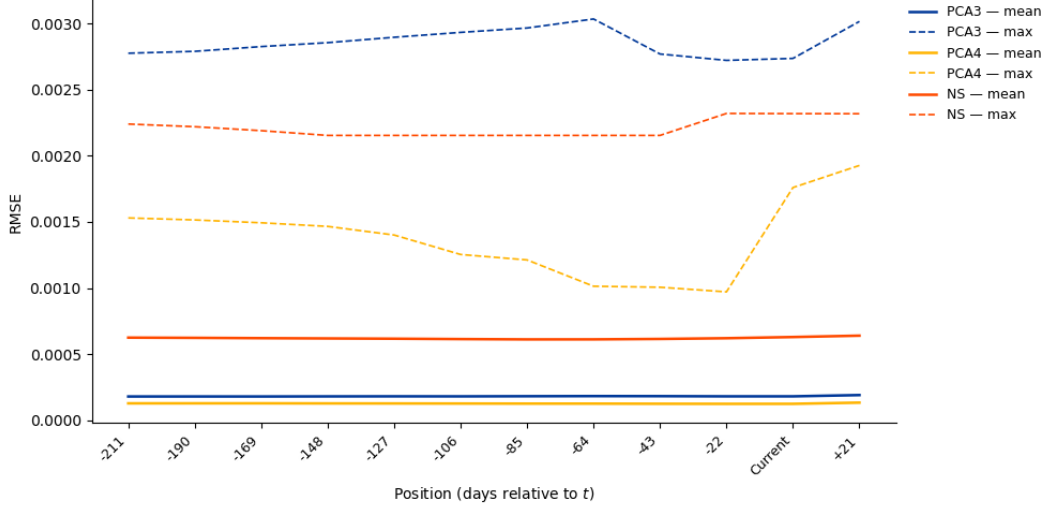


Figure 4: Mean and maximum reconstruction RMSE by position for PCA3, PCA4 and NS, over 6209 observations.

Figure 4 reports the mean and maximum reconstruction RMSE across all 6209 observations, from position -211 to $+21$ trading days. The mean lines are virtually flat across all positions for all three methods, indicating that on average the stale loading structure represents the yield curve equally well at past positions and at the forecast horizon. The maximum lines, reflecting the worst cases in the distribution, are similarly stable across past positions but show a marginal increase at $+21$ for both PCA specifications, visible also in the distributional results of Appendix A1. This marginal deterioration in the tail is expected, as the loading matrix is estimated on data up to t and is applied one month into the future. Crucially, the mean is unaffected, confirming that assumptions (15) and (6) hold in expectation. Among the PCA specifications, PCA4 exhibits a more stable distribution of fitting errors under the loadings assumption than PCA3 across all positions, with consistently lower RMSE indicating that the four-component basis represents yield curve changes more reliably at every position.

4.7 Reconstruction Assessment

Having validated loading stability, we assess how well each method reconstructs the actual yield curve at the forecast horizon $t+h$. Using stale loadings estimated at t , the reconstructed yield curve at $t+h$ follows (14) for PCA and (7) for NS. Reconstruction errors are computed as per-tenor absolute deviations against actual yields at $t+h$, across all 6,209 observations.

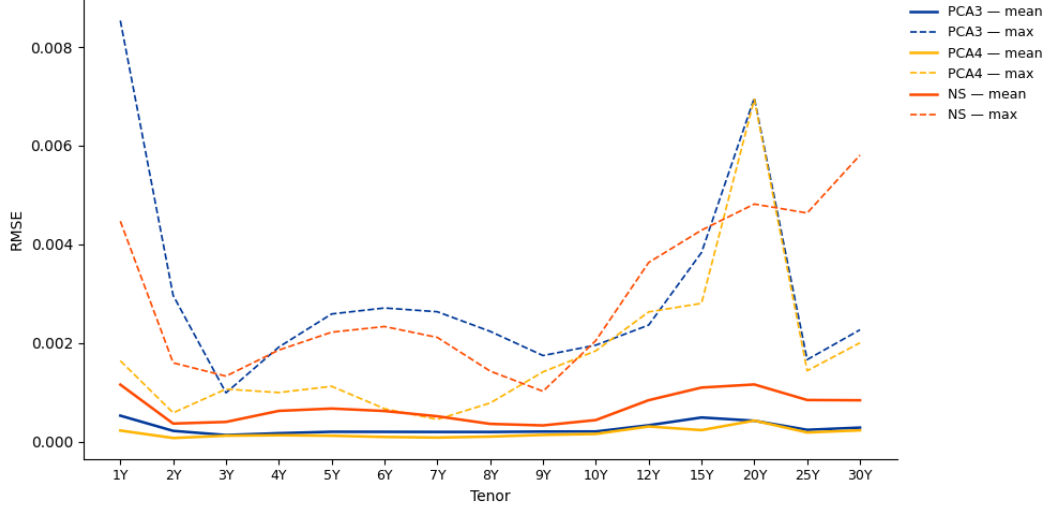


Figure 5: Mean and maximum reconstruction RMSE by tenor for PCA3, PCA4 and NS, over 6,209 observations.

Figure 5 shows mean and maximum reconstruction RMSE across tenors. PCA4 achieves the lowest mean errors at every tenor, followed closely by PCA3, with NS performing considerably worse throughout. The maximum lines reflect isolated worst-case observations rather than systematic behaviour. Detailed distributional results by tenor are provided in Appendix A2, which also presents reconstruction RMSE per tenor over time; stress episodes such as the 2008 financial crisis, the 2012 sovereign debt crisis, and the 2020 Covid shock produce the largest deviations across all three methods, though PCA3 and PCA4 remain at considerably lower magnitudes than NS throughout.

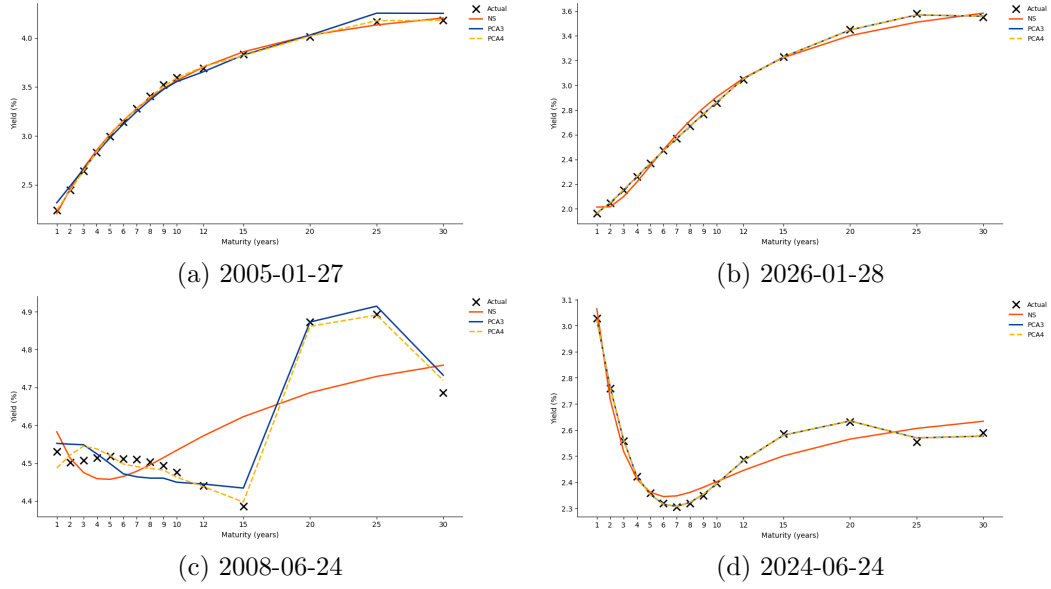


Figure 6: Yield curve fits for selected dates.

Figure 6 illustrates the fit across four yield curve environments. NS fails to capture irregular shapes such as the 2008 hump and the 2024 inverted curve, while PCA4 tracks the actual curve closely in all four cases. These findings confirm that PCA4 provides the most reliable reconstruction across tenors, time periods, and yield curve shapes, and motivate its use as the basis for the forecasting framework developed in subsequent sections. Within this comparison, NS serves as the natural benchmark given its role in the Diebold and Li (2006) framework this thesis extends, and PCA4's mean reconstruction error is two to five times smaller than NS across tenors (Appendix A2). Whether reconstruction errors of this magnitude are material for the forecasting results that follow is a separate question, addressed in Section 7 once out-of-sample forecast errors are available for comparison.

5 Forecasting Principal Components with VAR

5.1 VAR Specification and In-Sample Results

The autocorrelation analysis of Section 4.4 motivates a vector autoregression (VAR) over the four principal component changes $\Delta \mathbf{f}_t = (\Delta f_{1,t}, \Delta f_{2,t}, \Delta f_{3,t}, \Delta f_{4,t})^\top$. Following Diebold and Li (2006), who establish the VAR as the natural benchmark for factor-based yield curve forecasting, we model each component jointly with the others using the forecastable lag set $\mathcal{L}^*$. The system is:

$$\Delta f_{k,t} = c_k + \sum_{\ell \in \mathcal{L}^*} \sum_{j=1}^4 \phi_{k,j}^{(\ell)} \Delta f_{j,t-\ell} + \varepsilon_{k,t}, \quad k \in \{1, 2, 3, 4\} \quad (21)$$

where c_k is an intercept, $\phi_{k,j}^{(\ell)}$ is the coefficient on the ℓ -th lag of component j in equation k , and $\varepsilon_{k,t}$ is the equation error. Each equation contains $|\mathcal{L}^*| \times 4 = 40$ regressors and is estimated separately by OLS:

$$\hat{\boldsymbol{\theta}}_k = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}_k^* \quad (22)$$

where $\mathbf{X}$ is the $N \times 41$ design matrix of lagged regressors plus intercept and $\mathbf{y}_k^* \in \mathbb{R}^N$ is the vector of observations of $\Delta f_{k,t}$. Table 3 reports the in-sample fit statistics. The in-sample RMSE is the root mean squared residual:

$$\text{RMSE}_k^{\text{IS}} = \sqrt{\frac{1}{N} \sum_{t=1}^N \hat{\varepsilon}_{k,t}^2} \quad (23)$$

The in-sample R^2 is the standard coefficient of determination:

$$R_k^2 = 1 - \frac{\sum_{t=1}^N \hat{\varepsilon}_{k,t}^2}{\sum_{t=1}^N (\Delta f_{k,t} - \overline{\Delta f_k})^2} \quad (24)$$

where $\overline{\Delta f_k}$ is the in-sample mean. AIC and BIC are the Akaike and Bayesian information criteria computed from the Gaussian log-likelihood with $p = 40$ regressors plus intercept; lower values indicate a better penalised fit. DoF denotes residual degrees of freedom $N - p - 1 = 5,957$.

Table 3: VAR in-sample fit statistics, $N = 5,998$.

$\Delta f_{k,t}$	N	R^2	RMSE	AIC	BIC	DoF
$\Delta f_{1,t}$	5998	0.4683	0.007534	-41537.15	-41262.49	5957
$\Delta f_{2,t}$	5998	0.4668	0.002498	-54779.91	-54505.24	5957
$\Delta f_{3,t}$	5998	0.3362	0.001242	-63163.09	-62888.42	5957
$\Delta f_{4,t}$	5998	0.4581	0.000867	-67470.89	-67196.22	5957

All four equations achieve substantial explanatory power, with R^2 ranging from 0.34 for Δf_3 to 0.47 for Δf_1 and Δf_2 , confirming that the lag structure identified from the ACF and PACF carries genuine predictive content. Full per-equation coefficient tables with standard errors, t -statistics, and significance levels are provided in Appendix A4.

Table 4: Cross-PC significance matrix. Each entry reports the number of statistically significant lags out of 10 at $\alpha = 0.05$.

Equation \ Predictor	$\Delta f_{1,t}$	$\Delta f_{2,t}$	$\Delta f_{3,t}$	$\Delta f_{4,t}$
$\Delta f_{1,t}$	10/10	7/10	9/10	7/10
$\Delta f_{2,t}$	8/10	8/10	8/10	8/10
$\Delta f_{3,t}$	8/10	4/10	7/10	8/10
$\Delta f_{4,t}$	9/10	8/10	6/10	8/10

Table 4 summarises the number of statistically significant cross-lags per predictor-equation pair. The principal components, while orthogonal by construction in the cross-sectional sense, are not dynamically independent. Lagged values of Δf_1 , the parallel shift factor carrying the largest share of yield curve variance, are significant in at least 8 out of 10 lags in every non-own equation, making it the strongest cross-predictor in the system. This is economically interpretable: parallel level shifts propagate into slope and curvature dynamics with a delay. The implication is important for model specification: a univariate autoregressive model for each component would discard this cross-predictive information, and the VAR framework is therefore the appropriate linear benchmark.

5.2 Rolling VAR and Out-of-Sample Results

To eliminate look-ahead bias, we implement a rolling walk-forward procedure. At each prediction date t , the VAR is estimated on the τ most recent observations ending at $t - h$ and used to predict $\Delta f_{k,t}$:

$$\widehat{\Delta f}_{k,t} = \hat{c}_k^{(t)} + \sum_{\ell \in \mathcal{L}^*} \sum_{j=1}^4 \hat{\phi}_{k,j}^{(\ell,t)} \Delta f_{j,t-\ell} \quad (25)$$

where the superscript (t) denotes that coefficients are re-estimated at each step using only data in $[t - h - \tau, t - h]$. The directional prediction is:

$$\hat{d}_{k,t} = \mathbb{1} \left[\widehat{\Delta f}_{k,t} > 0 \right] \quad (26)$$

evaluated against the realised direction $d_{k,t} = \mathbb{1}[\Delta f_{k,t} > 0]$. Directional accuracy is then defined as:

$$\text{DA}_k = \frac{1}{N} \sum_{t=1}^N \mathbb{1} \left[\hat{d}_{k,t} = d_{k,t} \right] \quad (27)$$

Out-of-sample forecast performance is measured by RMSE and the out-of-sample R^2 . The out-of-sample RMSE for component k is:

$$\text{RMSE}_k = \sqrt{\frac{1}{N} \sum_{t=1}^N \left(\Delta f_{k,t} - \widehat{\Delta f}_{k,t} \right)^2} \quad (28)$$

The out-of-sample R^2 is defined following Campbell (2008) as:

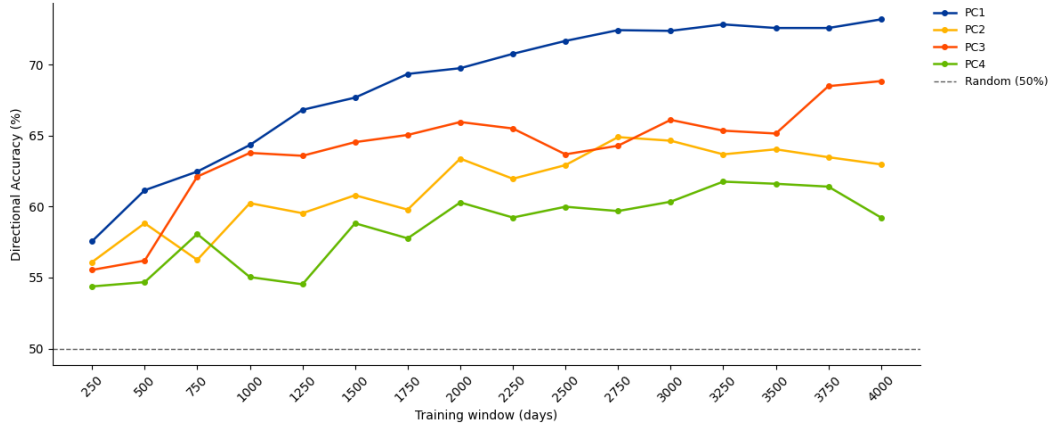
$$R_{\text{OOS},k}^2 = 1 - \frac{\sum_{t=1}^N \left(\Delta f_{k,t} - \widehat{\Delta f}_{k,t} \right)^2}{\sum_{t=1}^N \left(\Delta f_{k,t} - \overline{\Delta f}_k \right)^2} \quad (29)$$

where $\overline{\Delta f_k}$ is the historical mean of $\Delta f_{k,t}$ over the evaluation window. This measure compares the forecast model against a simple mean benchmark: $R_{\text{OOS}}^2 > 0$ indicates that the model beats the historical mean in terms of mean squared prediction error, while $R_{\text{OOS}}^2 < 0$ signals that the model underperforms the mean benchmark, which may indicate overfitting of the training window to in-sample dynamics.

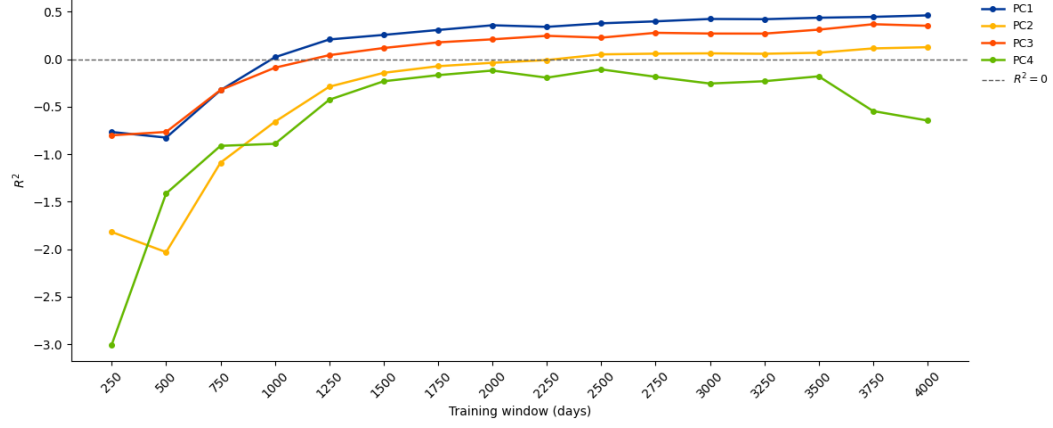
5.2.1 Training Window Selection

The choice of τ involves a trade-off between parameter stability and estimation precision: longer windows reduce parameter uncertainty across 40 regressors per equation but may incorporate observations that are no longer representative. We evaluate candidate windows $\tau \in \{250, 500, \dots, 4000\}$ on a common evaluation sample of $n = 1,977$ observations from 2018-06-22 to 2026-03-13.

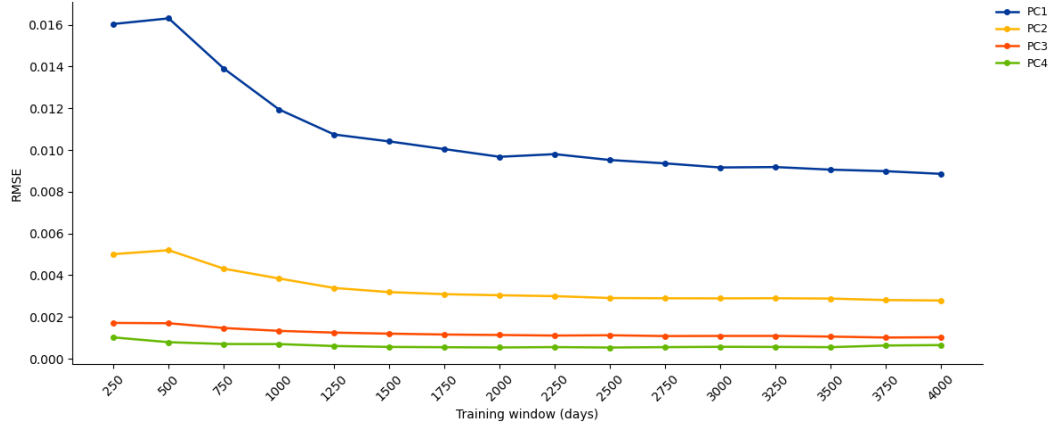
Figure 7 presents the results. Three patterns emerge. First, directional accuracy increases monotonically with τ for Δf_1 , stabilising around $\tau = 2,500$, while improvements for Δf_2 , Δf_3 , and Δf_4 are less systematic beyond $\tau = 2,000$. Second, the out-of-sample R^2 (29) is deeply negative at small windows, indicating that the model overfits the short training sample and produces predictions worse than the historical mean. It converges to positive values from $\tau \approx 1,250$ for Δf_1 and Δf_3 , while remaining negative for Δf_4 throughout. Third, RMSE (28) decreases steadily with τ for all components up to approximately $\tau = 3,000$, with only marginal improvements at larger windows. We select $\tau = 3,000$ as a defensible choice for our forecasting framework: accuracy and RMSE are near their best values across components, while retaining $N = 2,977$ out-of-sample predictions. Extending to $\tau = 4,000$ yields negligible further improvement in accuracy or RMSE while substantially reducing the evaluation horizon.



(a) Directional Accuracy



(b) Out-of-Sample R^2



(c) RMSE

Figure 7: Rolling VAR directional accuracy, out-of-sample R^2 (29) and RMSE (28) as a function of training window size τ , evaluated on the common sample ($n = 1,977$).

5.2.2 Out-of-Sample Results

A notable finding is that Δf_3 achieves both higher directional accuracy (64.43%) and substantially higher out-of-sample R^2 (0.263) than Δf_2 (65.87%, 0.053), despite Δf_2 explaining more yield curve variance in-sample: curvature changes appear more linearly predictable in magnitude than slope changes. For Δf_4 the VAR achieves 62.08% directional accuracy, though the negative out-of-sample R^2 of -0.21 indicates that the magnitude of Δf_4 changes is difficult to forecast linearly, even where direction is captured. Residual means across all components are negligibly close to zero, confirming the absence of systematic bias. The VAR thus establishes the linear benchmark for yield curve factor forecasting. The following section investigates whether an LSTM can capture non-linear dynamics in the factor series and improve upon this baseline.

Table 5: Out-of-Sample Rolling VAR, $\tau = 3,000$.

$\Delta f_{k,t}$	N	Accuracy	R^2	RMSE	AIC	BIC
$\Delta f_{1,t}$	2977	72.09%	0.4092	0.008325	-19980.27	-19734.33
$\Delta f_{2,t}$	2977	65.87%	0.0532	0.002730	-26619.07	-26373.13
$\Delta f_{3,t}$	2977	64.43%	0.2628	0.001032	-32411.67	-32165.72
$\Delta f_{4,t}$	2977	62.08%	-0.2065	0.000567	-35974.91	-35728.97

Table 6: Out-of-Sample Rolling VAR forecast residual statistics, $\tau = 3,000$.

$\Delta f_{k,t}$	Mean	Std. Dev.	Min	Q25	Q75	Max
$\Delta f_{1,t}$	0.000158	0.008324	-0.036123	-0.004629	0.004874	0.036659
$\Delta f_{2,t}$	0.000046	0.002729	-0.012505	-0.001365	0.001233	0.015597
$\Delta f_{3,t}$	0.000037	0.001031	-0.004325	-0.000547	0.000644	0.005226
$\Delta f_{4,t}$	0.000036	0.000566	-0.002095	-0.000271	0.000317	0.003605

6 Forecasting Principal Components with LSTM

6.1 Feature Construction

Two categories of input features are distinguished: sequential features, which vary across timesteps and are processed by the recurrent layers, and static features, which are constant across timesteps and concatenated after the recurrent pass.

6.1.1 Sequential Features

For each $\ell \in \mathcal{L}$, a ten-dimensional feature vector is constructed. Four features are the factor coordinates $\mathbf{f}_{t-\ell}$ as defined in (18), and one is the reconstruction error $\text{RMSE}_{t-\ell}$ as defined in (19); both were introduced in Section 4.5.

The 3M implied swaption volatility, expressed in basis points, is transformed into a 21-day percentage return series, $\Delta \text{IV}_{t-\ell} = (\text{IV}_{t-\ell} - \text{IV}_{t-\ell-21}) / \text{IV}_{t-\ell-21}$, and used as a sixth sequential feature in LSTM-VAR-FULL-EIG-IV only. This provides a signal about market-implied rate uncertainty at each lag position. We take the difference as Δf_t represents the change in momentum, which the change in IV should also represent.

The remaining four features are the lagged factor differences $\Delta f_{k,t-\ell}$ for $k \in \{1, 2, 3, 4\}$, as defined in (16). At $\ell = 0$ these are set to zero:

$$\Delta f_{k,t} = 0, \quad \ell = 0 \quad (30)$$

since $\Delta f_{k,t} = f_{k,t+h} - f_{k,t}$ involves the forecast target and is unavailable at prediction time. The sequential input is thus a tensor of shape $11 \times p$, where $p \in \{4, 5, 9, 10\}$ depending on the variant; see Table 7.

6.1.2 Static Features

The four daily eigenvalues $\boldsymbol{\nu}_t^\delta = (\nu_{1,t}^\delta, \dots, \nu_{4,t}^\delta)$ from $\hat{\boldsymbol{\Sigma}}_t^\delta$, as introduced in (8), are fed as static context. These capture the time-varying explained variance per component and provide a compact summary of the prevailing yield curve volatility regime.

6.1.3 Feature Standardisation

All features are standardised by a **StandardScaler** fitted on the training window $[t-h-\tau, t-h)$ and applied to the prediction observation. Concretely, for a feature with in-window mean μ and standard deviation σ :

$$\tilde{x} = \frac{x - \mu}{\sigma} \quad (31)$$

Fitting exclusively on the training window prevents data leakage: using any information from t or beyond would implicitly reveal the distributional location of the quantity being forecast. Rolling re-standardisation also stabilises the input scale across volatility regimes (Goodfellow et al., 2016).

6.2 Model Variants

Five variants are evaluated. Table 7 summarises their feature sets and architectures; Figure 8 shows the full graph of the largest variant.

Table 7: LSTM variant feature sets and architectures. $\mathbf{F}_{t-\ell} = (\mathbf{f}_{t-\ell}, \text{RMSE}_{t-\ell})$, $\mathbf{G}_{t-\ell} = \Delta \mathbf{f}_{k,t-\ell}$.

Variant	Sequential features	p	Static	Head
LSTM-VAR	$\mathbf{G}_{t-\ell}$	4	—	Dense(4)
LSTM-FULL	$\mathbf{F}_{t-\ell}$	5	—	Dense(4)
LSTM-FULL-EIG	$\mathbf{F}_{t-\ell}$	5	$\boldsymbol{\nu}_t^\delta$	Dense(16) + Dense(4)
LSTM-VAR-FULL-EIG	$\mathbf{F}_{t-\ell}, \mathbf{G}_{t-\ell}$	9	$\boldsymbol{\nu}_t^\delta$	Dense(16) + Dense(4)
LSTM-VAR-FULL-EIG-IV	$\mathbf{F}_{t-\ell}, \mathbf{G}_{t-\ell}, \text{IV}_{t-\ell}$	10	$\boldsymbol{\nu}_t^\delta$	Dense(16) + Dense(4)

All variants share the recurrent backbone: two stacked LSTM layers of 50 and 20 units. The literature provides no strong consensus on whether stacked LSTM architectures offer a systematic advantage over single-layer architectures for financial time series forecasting. Greff et al. (2015), in a large-scale study of LSTM variants, found that increased architectural complexity did not significantly improve performance over a single-layer LSTM with 20–200 units; accordingly, Nunes et al. (2026) adopt a single-layer LSTM with 100 units for bond yield forecasting. Ayitey Junior et al. (2022) compare several recurrent architectures, including a stacked LSTM, GRU, and bidirectional LSTM, on forex rate forecasting, and find that the plain recurrent baselines are themselves broadly comparable, with the strongest and most consistent gains instead attributable to an added attention mechanism rather than to depth. Taken together, the evidence suggests that architectural variation within the LSTM family, including the choice between a single layer and a modest stack, is unlikely to be the binding constraint on performance, while a more targeted mechanism for filtering information may matter more; this thesis accordingly retains a standard two-layer stack and instead varies the input representation across variants, as motivated in Section 6.

LSTM-VAR uses only $\Delta \mathbf{f}_{k,t-\ell}$, the same information as the VAR in (21), making it the direct nonlinear benchmark. LSTM-FULL replaces these with the coordinate-consistent factor representation from Section 4.5, adding $\text{RMSE}_{t-\ell}$ as a quality signal. LSTM-FULL-EIG augments this with the eigenvalues as static regime context. LSTM-VAR-FULL-EIG-IV combines all information sources, adding both $\Delta \mathbf{f}_{k,t-\ell}$ and $\Delta \text{IV}_{t-\ell}$ to the sequential input.

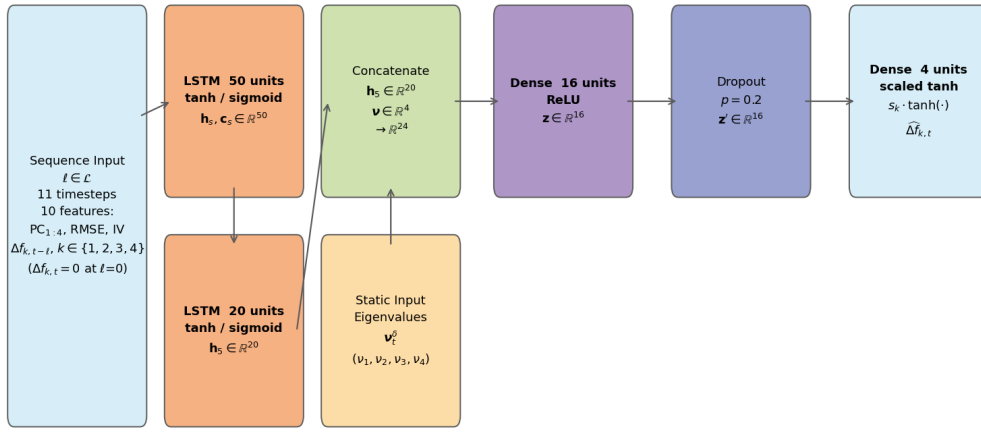


Figure 8: Architecture of LSTM-VAR-FULL-EIG-IV. Variants without static input omit the concatenation and Dense(16) head; see Table 7.

6.3 Architecture and Training

6.3.1 LSTM Layers

At each timestep $s \in \{1, \dots, S\}$, with $S = 11$, the LSTM updates its hidden state $\mathbf{h}_s \in \mathbb{R}^d$ and cell state $\mathbf{c}_s \in \mathbb{R}^d$ following Hochreiter and Schmidhuber (1997) via:

$$\mathbf{f}_s = \sigma(\mathbf{W}_f[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_f) \quad (32)$$

$$\mathbf{i}_s = \sigma(\mathbf{W}_i[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_i) \quad (33)$$

$$\tilde{\mathbf{c}}_s = \tanh(\mathbf{W}_c[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_c) \quad (34)$$

$$\mathbf{c}_s = \mathbf{f}_s \odot \mathbf{c}_{s-1} + \mathbf{i}_s \odot \tilde{\mathbf{c}}_s \quad (35)$$

$$\mathbf{o}_s = \sigma(\mathbf{W}_o[\mathbf{h}_{s-1}, \mathbf{x}_s] + \mathbf{b}_o) \quad (36)$$

$$\mathbf{h}_s = \mathbf{o}_s \odot \tanh(\mathbf{c}_s) \quad (37)$$

where σ denotes the sigmoid function and $\odot$ elementwise multiplication. The sigmoid gates and tanh activations are the original formulation of Hochreiter and Schmidhuber (1997): the sigmoid constrains gate outputs to $(0, 1)$, acting as a soft switch, while tanh bounds cell state and hidden state values to $(-1, 1)$, providing a natural normalisation for the recurrent activations. The forget gate $\mathbf{f}_s$ controls how much of the previous cell state is retained, the input gate $\mathbf{i}_s$ determines what new information enters, and the output gate $\mathbf{o}_s$ filters what is passed forward as the hidden state. The cell state $\mathbf{c}_s$ therefore acts as a selective memory that can propagate information across the full lag horizon without the vanishing gradient problem of plain RNNs.

Two LSTM layers are stacked: the first with $d = 50$ units processes the raw temporal patterns across the 11 timesteps, the second with $d = 20$ units compresses this into a lower-dimensional representation before the feedforward head. All states are initialised at zero, $\mathbf{h}_0 = \mathbf{c}_0 = \mathbf{0}$, since no prior information about the sequence is available at the start of each window.

6.3.2 Feedforward Head

For variants without static input (LSTM-VAR, LSTM-FULL), the final hidden state $\mathbf{h}_S \in \mathbb{R}^{20}$ is passed directly to the output layer. For variants with static input (LSTM-FULL-EIG, LSTM-VAR-FULL-EIG-IV), $\mathbf{h}_S$ is concatenated with the standardised eigenvalues $\tilde{\boldsymbol{\nu}}_t^\delta \in \mathbb{R}^4$ to form a vector in $\mathbb{R}^{24}$, which is passed through a ReLU activation (Goodfellow et al., 2016):

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_1[\mathbf{h}_S, \tilde{\boldsymbol{\nu}}_t^\delta] + \mathbf{b}_1), \quad \text{ReLU}(x) = \max(0, x) \quad (38)$$

ReLU is preferred over sigmoid or tanh in the feedforward head as it avoids saturation in non-recurrent layers and has been shown to accelerate convergence in deep networks (Goodfellow et al., 2016). This is followed by Dropout with $p = 0.2$ (Srivastava et al., 2014), which randomly zeroes activations during training to reduce overfitting.

6.3.3 Output Layer and Scaled Tanh

All four components are predicted simultaneously. The raw output of the final Dense(4) layer is passed through a scaled tanh:

$$\widehat{\Delta f}_{k,t+h} = s_k \cdot \tanh(\mathbf{w}_k^\top \mathbf{z}'), \quad s_k = \max\left(\max_t \Delta f_{k,t}, -\min_t \Delta f_{k,t}\right) \quad (39)$$

where $\mathbf{z}'$ denotes the penultimate activation and s_k is the maximum absolute value of $\Delta f_{k,t}$ over the training window $[t-h-\tau, t-h]$. The targets are normalised to $[-1, 1]$ before training via $\Delta f_{k,t}/s_k$. This bounds predictions within the observed range of factor movements, prevents extrapolation beyond historically observed dynamics, and stabilises training when factor magnitudes vary across rolling windows. The scaling is recomputed at each rolling step.

6.3.4 Training Procedure

All variants are estimated using a rolling walk-forward procedure identical to that of the VAR in Section 5.2. At each prediction date t , a fresh model is initialised and trained from scratch on the $\tau = 3,000$ most recent observations in $[t-h-\tau, t-h]$, then used to produce a single out-of-sample prediction for $\widehat{\Delta f}_{k,t+h}$. No parameters are carried forward across windows.

Each rolling step requires a full training run, and with $N = 2,977$ prediction dates the total computation amounts to thousands of independent training jobs. The models were implemented in TensorFlow and executed on a 16-core, 64 GB Cloudera AI environment with 16 parallel workers, each responsible for an independent subset of rolling windows. This design is nonetheless the natural evaluation framework: it mirrors exactly how a practitioner would deploy the model, with access to the most recent 3,000 observations and a requirement to produce one forecast for the next trading month.

The loss function is mean absolute error between the scaled predictions and the normalised targets $\Delta f_{k,t}/s_k$. Training uses the Adam optimiser (Kingma and Ba, 2017) with learning rate $\alpha = 0.001$, batch size 32, and a maximum of 500 epochs. A `ReduceLROnPlateau` callback halves the learning rate after 10 epochs without improvement in training loss, with a floor of 10^{-5} ; learning rate scheduling and early stopping are standard regularisation techniques for deep networks (Goodfellow et al., 2016). Early stopping with patience 40 and `restore_best_weights=True` terminates training when training loss ceases to improve, returning the lowest-loss checkpoint. Since the model is evaluated entirely out-of-sample, the training window $[t-h-\tau, t-h]$ never overlaps with the prediction date t , and the rolling design itself provides the holdout.

6.4 Out-of-Sample Results

Table 8 reports directional accuracy, out-of-sample R^2 (29), and RMSE (28) for all variants alongside the VAR benchmark from Table 5. Full per-variant performance and residual tables are provided in Appendix A5.

Table 8: Out-of-sample comparison: VAR and LSTM variants, $\tau = 3,000$, $N = 2,977$.

Variant	Accuracy (%)				R^2				RMSE ($\times 10^3$)			
	PC1	PC2	PC3	PC4	PC1	PC2	PC3	PC4	PC1	PC2	PC3	PC4
VAR	72.09	65.87	64.43	62.08	0.409	0.053	0.263	-0.207	8.33	2.73	1.03	0.57
LSTM-VAR	64.70	56.73	59.05	57.10	0.095	-0.248	-0.155	-0.657	10.30	3.13	1.29	0.67
LSTM-FULL	70.27	68.26	65.97	65.50	0.236	0.158	0.096	-0.129	9.47	2.58	1.14	0.55
LSTM-FULL-EIG	69.70	65.57	65.77	55.59	0.257	0.137	0.176	-0.112	9.34	2.61	1.09	0.54
LSTM-VAR-FULL-EIG	71.35	66.78	66.31	58.92	0.295	0.178	0.192	-0.116	9.09	2.54	1.08	0.55
LSTM-VAR-FULL-EIG-IV	71.08	67.48	65.40	60.36	0.318	0.225	0.198	0.003	8.95	2.47	1.08	0.52

6.4.1 LSTM-VAR versus VAR

LSTM-VAR operates on identical information to the VAR and performs worse on all three metrics across every component. PC1 accuracy falls from 72.09% to 64.70%, R^2 from 0.41 to 0.09, and RMSE ($\times 10^3$) widens from 8.33 to 10.30. The negative R^2 values for PC2 through PC4 indicate that the model’s continuous predictions are worse than a simple mean forecast in terms of magnitude, even where directional accuracy remains above 50%. The mean-reversion structure identified in Section 4.4 is well captured linearly by the VAR, and the specific LSTM architecture chosen here does not replicate that structure when restricted to the same inputs.

A likely contributing factor is the severe disparity between model complexity and available training data. The VAR contains $4 \times (40 + 1) = 164$ parameters estimated on $\tau = 3,000$ observations, while LSTM-VAR contains approximately 16,800 parameters across its recurrent and output layers, a ratio of roughly 100:1 relative to the VAR, and less than 0.4 observations per parameter even across the full dataset of 6,209 observations. Under these conditions the LSTM is unlikely to reliably identify the lag structure from the available data, which may explain the underperformance relative to the VAR. This data-to-parameter imbalance is a known challenge when applying LSTMs to financial time series, where training samples are typically limited relative to the model’s capacity (Fischer and Krauss, 2018).

6.4.2 LSTM-FULL versus LSTM-VAR and VAR

Replacing the raw factor differences with the coordinate-consistent representation as introduced in section 4.5 substantially improves all metrics. Directional accuracy exceeds the VAR for PC2 (68.26% vs 65.87%), PC3 (65.97% vs 64.43%), and PC4 (65.50% vs 62.08%), and RMSE improves for PC2 and PC4 relative to both the VAR and LSTM-VAR. The R^2 for PC2 rises from 0.053 to 0.158, a meaningful improvement in magnitude prediction, while PC4 remains negative but improves from -0.207 under the VAR to -0.129 . PC1 remains below the VAR on both accuracy and R^2 , consistent with the level factor being the most linearly predictable component.

The gains for PC2 and PC4 in particular suggest that nonlinear dynamics may be present for these factors that the VAR cannot capture. For PC4, the R^2 remaining negative indicates that magnitude prediction is still difficult. Taken together, LSTM-FULL provides evidence that the coordinate-consistent representation from Section 4.5 gives the LSTM access to genuinely incremental information.

6.4.3 LSTM-FULL-EIG versus LSTM-FULL

Adding eigenvalues as static context produces mixed results. R^2 improves for PC1 (0.236 to 0.257) and PC3 (0.096 to 0.176), and RMSE is marginally lower across most components. However, directional accuracy declines for PC2 (68.26% to 65.57%) and PC4 (65.50% to 55.59%), with the latter dropping below the VAR benchmark. Where accuracy falls while R^2 and RMSE improve only marginally, the model sacrifices directional correctness without a compensating gain in magnitude prediction. The eigenvalues do not add consistent value across components.

6.4.4 Combined Variant

LSTM-VAR-FULL-EIG-IV achieves the strongest results across the magnitude metrics. It records the best R^2 for PC2 (0.225) and PC4 (0.003), and the lowest RMSE for PC2 (2.47) and PC4 (0.52). Notably, PC4 is the only component where any variant achieves a positive R^2 , indicating that combining all information sources finally enables the model to predict the magnitude of higher-order factor movements above the mean forecast baseline. On PC1, it closely approaches the VAR in both accuracy (71.08% vs 72.09%) and RMSE (8.95 vs 8.33), though the VAR retains the best R^2 for the level factor.

Comparing LSTM-VAR-FULL-EIG-IV to its no-IV counterpart LSTM-VAR-FULL-EIG reveals that the contribution of implied volatility is modest. The two variants perform similarly across most components; however, the with-IV variant achieves lower RMSE for all PCs, and is the only specification for which the R^2 of PC4 is non-negative (0.003 vs -0.116). The marginal improvement suggests that swaption implied volatility carries a small incremental signal for higher-order factor dynamics, though its effect is concentrated in PC4 and does not generalise uniformly across components.

Two strands of the literature offer candidate explanations for why this benefit is concentrated rather than uniform. Nunes et al. (2026) show, through direct inspection of a trained LSTM's internal gating signals, that individual hidden units specialise around specific exogenous variables and lag structures rather than each unit drawing on all available information equally; under this view, the IV feature may simply be exploited by a subset of units whose influence happens to concentrate on PC4, rather than being uniformly uninformative for the others. Separately, Bie et al. (2024) document that the predictive content of macroeconomic variables for the yield curve is itself regime-dependent, holding in some periods and not others; if the relevance of swaption implied volatility is similarly conditional on the prevailing rate or volatility regime, an evaluation window spanning multiple regimes would show benefits concentrated in particular periods or components without this implying the variable is uninformative in general. The current evaluation addresses input relevance only indirectly, through the progression of variants from LSTM-VAR through LSTM-FULL, LSTM-FULL-EIG, and LSTM-VAR-FULL-EIG-IV, each adding one feature group at a time, with the marginal change in performance across variants serving as a proxy for feature importance. Distinguishing between the two candidate explanations above directly, for instance through the gate-level inspection of Nunes et al. (2026), is left for future work.

Directional accuracy is more mixed: LSTM-FULL retains the highest accuracy for PC2 and PC4, suggesting that adding the raw factor differences and implied volatility improves magnitude prediction at some cost to directional sharpness. Across all metrics and components together, LSTM-VAR-FULL-EIG-IV represents the most balanced specification and will be carried forward to the yield curve reconstruction in Section 7.

7 Yield Curve Reconstruction

7.1 From Factor Forecasts to Yield Levels

The factor-level models developed in Sections 5 and 6 produce out-of-sample predictions $\widehat{\Delta f}_{k,t+h}$ for each component $k \in \{1, 2, 3, 4\}$ at horizon $h = 21$ trading days. To map these back to yield curve levels, it is necessary to clarify precisely what quantity the models predict.

The training target is defined as the change in the projected factor between consecutive 21-day windows, both expressed under the same loading matrix $\mathbf{L}_t$:

$$\Delta f_{k,t+h} = f_{k,t+h} - f_{k,t} = \mathbf{L}_t^T(\mathbf{y}_{t+h} - \mathbf{y}_t) - \mathbf{L}_t^T(\mathbf{y}_t - \mathbf{y}_{t-h}) \quad (40)$$

Collecting terms, this simplifies to the projection of the second difference of the yield curve:

$$\Delta f_{k,t+h} = \mathbf{L}_t^T(\mathbf{y}_{t+h} - 2\mathbf{y}_t + \mathbf{y}_{t-h}) \quad (41)$$

The models predict whether the yield curve will accelerate or decelerate relative to its current 21-day momentum. A positive prediction implies that yields will continue moving in the same direction faster than the current trend; a negative prediction implies deceleration or reversal.

The predicted factor level at $t + h$ is recovered by adding the current factor level back:

$$\widehat{\mathbf{f}}_{t+h} = \mathbf{f}_t + \widehat{\Delta \mathbf{f}}_{t+h} \quad (42)$$

where $\mathbf{f}_t = \mathbf{L}_t^T(\mathbf{y}_t - \mathbf{y}_{t-h})$ is the current 21-day factor move, known at prediction time. The predicted yield change is then obtained by projecting back through the loading matrix:

$$\widehat{\Delta \mathbf{y}}_{t+h} = \mathbf{L}_t \widehat{\mathbf{f}}_{t+h} = \mathbf{L}_t \mathbf{f}_t + \mathbf{L}_t \widehat{\Delta \mathbf{f}}_{t+h} \in \mathbb{R}^m \quad (43)$$

and the reconstructed yield curve level is:

$$\widehat{\mathbf{y}}_{t+h} = \mathbf{y}_t + \widehat{\Delta \mathbf{y}}_{t+h} = \mathbf{y}_t + \mathbf{L}_t \mathbf{f}_t + \mathbf{L}_t \widehat{\Delta \mathbf{f}}_{t+h} \quad (44)$$

The quality of this reconstruction depends on how well the k -dimensional subspace spanned by $\mathbf{L}_t$ captures yield curve variation, which is validated by the loading stability results of Section 4.6.

Four specifications are evaluated over the common out-of-sample window of $N = 2,977$ observations from 2014-08-22 to 2026-04-20: the rolling VAR, the LSTM-VAR-FULL-EIG-IV model, a hybrid model, and the random walk benchmark $\widehat{\mathbf{y}}_{t+h} = \mathbf{y}_t$.

The hybrid model assembles $\widehat{\Delta \mathbf{f}}_{t+h}$ from two sources, motivated by the factor-level results in Section 6.4. The VAR achieves the highest directional accuracy and R^2 for $\widehat{\Delta f}_{1,t+h}$ and $\widehat{\Delta f}_{3,t+h}$, the level and curvature components. The LSTM-VAR-FULL-EIG-IV model achieves superior R^2 and RMSE for $\widehat{\Delta f}_{2,t+h}$ and $\widehat{\Delta f}_{4,t+h}$, the slope and higher-order components. The hybrid prediction vector is therefore:

$$\widehat{\Delta \mathbf{f}}_{t+h}^{\text{hyb}} = \begin{pmatrix} \widehat{\Delta f}_{1,t+h}^{\text{VAR}} \\ \widehat{\Delta f}_{2,t+h}^{\text{LSTM}} \\ \widehat{\Delta f}_{3,t+h}^{\text{VAR}} \\ \widehat{\Delta f}_{4,t+h}^{\text{LSTM}} \end{pmatrix} \quad (45)$$

and the reconstruction follows (44) with $\widehat{\Delta \mathbf{f}}_{t+h}^{\text{hyb}}$ in place of the single-model predictions.

7.2 Unscaled Results

Table 9 reports RMSE and directional accuracy by tenor for all four specifications under the unscaled reconstruction (44).

Table 9: Unscaled yield curve reconstruction performance by tenor, $N = 2,977$.

Tenor	RMSE ($\times 10^3$)				Dir. Acc. (%)			
	RW	VAR	LSTM	Hybrid	RW	VAR	LSTM	Hybrid
1Y	1.705	1.813	1.969	1.829	0.03	52.91	51.29	52.70
2Y	1.774	1.899	2.095	1.948	0.03	53.85	50.45	52.30
3Y	1.845	1.998	2.211	2.059	0.10	53.24	49.75	52.40
4Y	1.895	2.078	2.297	2.140	0.03	52.40	49.41	51.66
5Y	1.927	2.132	2.351	2.188	0.00	52.33	49.61	51.53
6Y	1.950	2.170	2.381	2.212	0.10	53.54	50.55	51.83
7Y	1.969	2.207	2.403	2.231	0.10	53.01	50.86	52.40
8Y	1.992	2.255	2.431	2.257	0.03	51.86	50.86	52.57
9Y	2.016	2.308	2.462	2.288	0.07	52.17	51.36	52.74
10Y	2.031	2.344	2.482	2.307	0.03	51.76	51.29	52.03
12Y	2.091	2.465	2.552	2.389	0.00	51.02	52.67	52.10
15Y	2.135	2.507	2.582	2.429	0.03	50.72	52.40	52.00
20Y	2.197	2.612	2.647	2.514	0.00	50.69	52.50	51.90
25Y	2.177	2.637	2.645	2.520	0.00	49.98	52.67	51.56
30Y	2.179	2.637	2.646	2.522	0.03	52.23	51.76	50.82

The random walk achieves the lowest RMSE at every tenor. All three forecasting models produce substantially higher RMSE throughout, indicating that unscaled reconstruction overstates the magnitude of yield curve movements.

The directional picture is different. All three models achieve directional accuracy above 50% across most tenors, while the random walk registers near-zero accuracy by construction. The VAR leads at the short-to-mid curve, peaking at 53.85% at 2Y and 53.54% at 6Y. The LSTM underperforms at the short end, falling below 50% at 3Y–5Y, but recovers at longer tenors where it reaches 52.67% at 12Y and 25Y. The hybrid performs consistently across the curve, never dropping below 50%. The directional advantage across all models motivates the magnitude calibration procedure introduced in the following section. Full residual statistics for the unscaled reconstruction are provided in Appendix A6.1.

7.3 Magnitude Calibration via Alpha Scaling

The unscaled results confirm a consistent directional advantage but show that unit scaling substantially overstates the magnitude of predicted yield changes. This follows directly from (41): the models predict the second difference of the yield curve, and projecting back with $\alpha_1 = \alpha_2 = 1$ adds both the momentum term and the predicted acceleration at full scale,

amplifying the reconstruction beyond the realised yield change. A magnitude calibration seems as a logical next step.

The scaled reconstruction generalises (44) by introducing separate scaling coefficients for the two terms:

$$\hat{\mathbf{y}}_{t+h} = \mathbf{y}_t + \alpha_1 \cdot \mathbf{L}_t \mathbf{f}_t + \alpha_2 \cdot \mathbf{L}_t \widehat{\Delta \mathbf{f}}_{t+h} \quad (46)$$

The coefficients α_1 and α_2 are estimated jointly by OLS through the origin, regressing the realised yield change $\Delta \mathbf{y}_{t+h} = \mathbf{y}_{t+h} - \mathbf{y}_t$ on the two components across all $N \times m = 44,655$ observation-tenor pairs:

$$[\hat{\alpha}_1, \hat{\alpha}_2] =_{\alpha_1, \alpha_2} \sum_{i=1}^N \sum_{j=1}^m \left(\Delta y_{i,j} - \alpha_1 \cdot (\mathbf{L}_t \mathbf{f}_t)_{i,j} - \alpha_2 \cdot (\mathbf{L}_t \widehat{\Delta \mathbf{f}}_{t+h})_{i,j} \right)^2 \quad (47)$$

with closed-form solution $[\hat{\alpha}_1, \hat{\alpha}_2] = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ where the two columns of $\mathbf{X}$ are the flattened momentum and prediction terms respectively. If a perfectly calibrated model predicted the exact realised yield change, both α_1 and α_2 would equal one. The random walk corresponds to $\alpha_1 = \alpha_2 = 0$, since it predicts no yield change. The OLS regression therefore simultaneously tests whether the momentum term and the model predictions carry information beyond the random walk: a significant positive estimate for either coefficient is direct evidence that the corresponding term improves upon the no-change benchmark in magnitude. Values between zero and one indicate that the term points in the right direction but overstates the magnitude; negative values indicate that the term is directionally inverted relative to the realised change, which is economically incoherent and worse than the random walk.

7.3.1 In-Sample Estimates

Table 10 reports the in-sample OLS estimates for all three models, fitted over the full out-of-sample window of $N = 2,977$ observations.

Table 10: In-sample alpha estimates, $H_0: \alpha = 0$, $N \times m = 44,655$.

Model	$\hat{\alpha}_1$ (momentum)				$\hat{\alpha}_2$ (prediction)			
	Estimate	Std. Dev.	t	95% CI	Estimate	Std. Dev.	t	95% CI
VAR	0.0449	0.0081	5.54	[0.029, 0.061]	0.1367	0.0079	17.39	[0.121, 0.152]
LSTM-VAR-FULL-EIG-IV	-0.0896	0.0083	-10.80	[-0.106, -0.073]	-0.0210	0.0071	-2.95	[-0.035, -0.007]
Hybrid	0.0521	0.0084	6.24	[0.036, 0.068]	0.1421	0.0080	17.67	[0.126, 0.158]

The VAR and hybrid produce strictly positive and significant estimates for both $\hat{\alpha}_1$ and $\hat{\alpha}_2$, confirming that both the momentum term and the predicted acceleration carry genuine magnitude information in the correct direction. The LSTM produces negative estimates for both coefficients, of which $\hat{\alpha}_2$ is very close to zero, suggesting that the LSTM predictions do not carry meaningful magnitude and directional information at the yield curve level.

7.3.2 Stability Assessment

For the scaled reconstruction to be valid in a forward-looking context, $\hat{\alpha}_1$ and $\hat{\alpha}_2$ must remain stable and positive over time. A negative rolling alpha implies that the corresponding term is directionally inverting the prediction at that point in time, which is economically incoherent. The stability criterion is therefore simply:

$$\hat{\alpha}_k(t) > 0, \quad k \in \{1, 2\} \quad (48)$$

Both alphas are estimated on a rolling window of 2000 observations $[t - 2000 - 21, t - 21)$, where the upper bound $t - 21$ ensures that all training observations have forecast targets strictly before the prediction date t , so no future information enters the estimation. This window convention is shared with the out-of-sample procedure in Section 7.5. Full residual statistics for the in-sample scaled reconstruction are provided in Appendix A6.2. The rolling estimation yields 956 estimates over the period 2022-06-16 to 2026-03-13.

Figure 9 shows the rolling estimates for the VAR.

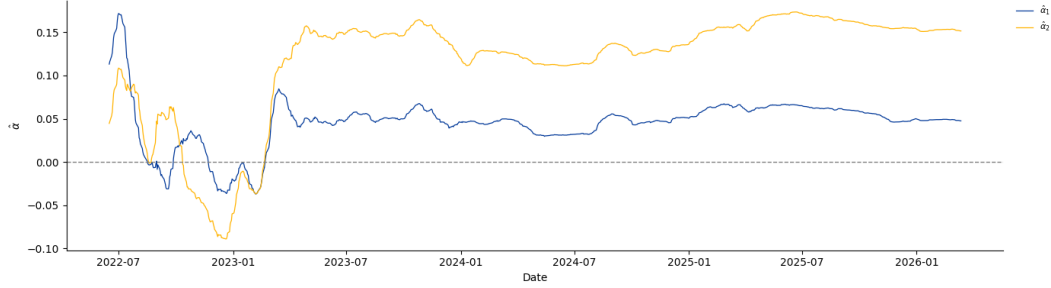


Figure 9: Rolling $\hat{\alpha}_1$ (blue) and $\hat{\alpha}_2$ (yellow) for the VAR. The dashed grey line marks zero; estimates below zero indicate an economically unstable scaling regime.

Both alphas are volatile and frequently negative in the initial period through late 2022, coinciding with the sharp ECB rate hiking cycle. From early 2023 onwards both estimates stabilise well above zero and remain positive for the bulk of the sample, with $\hat{\alpha}_2$ settling around 0.15 and $\hat{\alpha}_1$ around 0.05. Table 11 summarises the stability probabilities and distributional properties across all three models.

Table 11: Rolling alpha stability summary. Window = 2000 obs, 956 out-of-sample estimates, 2022-06-16 to 2026-03-13.

Model		Full-sample $\hat{\alpha}$	$P(\hat{\alpha} > 0)$	$P(\hat{\alpha} \leq 0)$	Mean	Std. Dev.
VAR	α_1	0.0449	90.27%	9.73%	0.0452	0.0289
	α_2	0.1367	89.96%	10.04%	0.1157	0.0628
LSTM-VAR-FULL-EIG-IV	α_1	-0.0896	6.69%	93.31%	-0.1250	0.0869
	α_2	-0.0210	7.74%	92.26%	-0.0737	0.0636
Hybrid	α_1	0.0521	92.99%	7.01%	0.0537	0.0295
	α_2	0.1421	90.48%	9.52%	0.1231	0.0605

The VAR and hybrid exhibit nearly identical stability profiles with $P(\hat{\alpha} > 0)$ around 90% for both alphas. The instability is concentrated in the 2022–2023 rate shock period and does not persist once the training window has absorbed the new rate regime. The LSTM presents a fundamentally different picture: both $\hat{\alpha}_1$ and $\hat{\alpha}_2$ are negative in the in-sample regression and remain negative in over 92% of rolling windows, confirming that the LSTM produces an economically unviable scaling throughout the evaluation period. The VAR and hybrid, by contrast, show stable and positive alphas in normal market environments, supporting the use of the two-alpha scaled reconstruction as a viable magnitude calibration for these models.

7.4 Scaled Results

Table 12 reports RMSE, R^2 , and directional accuracy for the VAR, LSTM, hybrid, and random walk under the two-alpha scaled reconstruction (46) with $\hat{\alpha}_1$ and $\hat{\alpha}_2$ estimated in-sample over the full $N = 2,977$ window.

Table 12: In-sample scaled yield curve reconstruction performance by tenor, $N = 2,977$.

Tenor	RMSE ($\times 10^3$)				R^2				Dir. Acc. (%)			
	RW	VAR	LSTM	Hybrid	RW	VAR	LSTM	Hybrid	RW	VAR	LSTM	Hybrid
1Y	1.705	1.689	1.724	1.693	0.000	0.019	-0.023	0.014	0.03	55.02	50.82	54.79
2Y	1.774	1.752	1.778	1.761	0.000	0.025	-0.005	0.015	0.03	56.50	51.90	54.62
3Y	1.845	1.819	1.839	1.830	0.000	0.027	0.006	0.015	0.10	56.73	53.07	54.99
4Y	1.895	1.869	1.884	1.880	0.000	0.027	0.011	0.016	0.03	56.43	53.07	53.75
5Y	1.927	1.902	1.914	1.912	0.000	0.026	0.013	0.015	0.00	55.32	52.17	53.95
6Y	1.950	1.928	1.937	1.935	0.000	0.022	0.013	0.014	0.10	54.99	51.53	53.75
7Y	1.969	1.951	1.957	1.956	0.000	0.019	0.012	0.013	0.10	54.72	51.33	53.78
8Y	1.992	1.978	1.981	1.980	0.000	0.014	0.011	0.012	0.03	53.98	52.00	53.71
9Y	2.016	2.005	2.006	2.004	0.000	0.011	0.010	0.012	0.07	53.44	52.60	53.41
10Y	2.031	2.023	2.022	2.020	0.000	0.008	0.008	0.011	0.03	53.61	52.44	53.91
12Y	2.091	2.090	2.085	2.083	0.000	0.001	0.006	0.008	0.00	53.01	52.30	53.95
15Y	2.135	2.133	2.133	2.126	0.000	0.002	0.003	0.009	0.03	53.31	52.64	53.61
20Y	2.197	2.201	2.197	2.190	0.000	-0.004	-0.001	0.006	0.00	51.93	52.07	53.14
25Y	2.177	2.180	2.176	2.167	0.000	-0.003	0.001	0.009	0.00	51.66	52.20	52.80
30Y	2.179	2.179	2.175	2.166	0.000	-0.000	0.003	0.012	0.03	52.23	52.47	53.21

Alpha scaling substantially improves both RMSE and R^2 relative to the unscaled results. The VAR beats the random walk on RMSE at 12 out of 15 tenors, with positive R^2 from 1Y to 15Y, peaking at 0.027 at 4Y–5Y. Directional accuracy ranges from 51.66% at 25Y to 56.73% at 3Y. The long end (20Y–30Y) shows marginally negative R^2 for the VAR, indicating that the magnitude calibration is less precise at very long maturities.

The hybrid beats the random walk on RMSE at all 15 tenors and achieves positive R^2 across the full curve, including the long end where the VAR turns marginally negative. Directional accuracy is consistently above 52% throughout and leads the VAR from 9Y onwards. As these results use in-sample alpha estimates, they are presented primarily to characterise the signal available in the factor predictions; the out-of-sample validation in Section 7.5 provides the honest assessment.

7.5 Out-of-Sample Validation

To provide a genuinely out-of-sample validation, we implement a rolling procedure in which both scaling coefficients are estimated exclusively on past data and applied only when the resulting estimates are economically stable.

At each prediction date t , $[\hat{\alpha}_1(t), \hat{\alpha}_2(t)]$ are estimated on the 2000 observations in $[t - 2000 - 21, t - 21]$. The upper bound $t - 21$ ensures that all training observations have forecast targets strictly before t , so no future yield information enters the alpha estimation. If either estimate is non-positive, no prediction is made for that date. Only dates where both $\hat{\alpha}_1(t) > 0$ and $\hat{\alpha}_2(t) > 0$ produce a forecast. The matched random walk is computed over exactly the same set of prediction dates, ensuring a like-for-like comparison.

Table 13 reports the results. Of the 956 candidate out-of-sample dates (2022-07-15 to

2026-04-20), the VAR uses 834 (87.2%) and the hybrid uses 853 (89.2%), with the remaining dates excluded due to non-positive alpha estimates concentrated in the 2022–2023 rate shock period. The mean $\hat{\alpha}_1$ and $\hat{\alpha}_2$ on the used dates are 0.053 and 0.137 for the VAR, and 0.060 and 0.141 for the hybrid, closely matching the in-sample estimates.

Table 13: Out-of-sample yield curve reconstruction performance by tenor. VAR: $N = 834$. Hybrid: $N = 853$. RW evaluated on matched dates.

Tenor	RMSE ($\times 10^3$)				R^2				Dir. Acc. (%)			
	RW _V	VAR	RW _H	Hybrid	RW _V	VAR	RW _H	Hybrid	RW _V	VAR	RW _H	Hybrid
1Y	2.190	2.185	2.466	2.470	0.000	0.005	0.000	−0.003	0.00	57.19	0.00	52.05
2Y	2.318	2.311	2.523	2.533	0.000	0.007	0.000	−0.008	0.00	56.47	0.00	51.82
3Y	2.386	2.377	2.558	2.569	0.000	0.008	0.000	−0.009	0.12	58.27	0.12	54.87
4Y	2.400	2.389	2.555	2.565	0.000	0.010	0.000	−0.008	0.12	57.67	0.12	54.63
5Y	2.374	2.361	2.521	2.529	0.000	0.011	0.000	−0.006	0.00	59.23	0.00	56.62
6Y	2.344	2.330	2.484	2.488	0.000	0.012	0.000	−0.004	0.12	59.95	0.12	56.86
7Y	2.315	2.302	2.447	2.449	0.000	0.011	0.000	−0.002	0.00	59.71	0.00	57.68
8Y	2.298	2.287	2.422	2.423	0.000	0.010	0.000	−0.000	0.12	59.59	0.12	57.91
9Y	2.287	2.277	2.402	2.401	0.000	0.009	0.000	0.001	0.00	59.47	0.00	58.15
10Y	2.265	2.257	2.376	2.375	0.000	0.007	0.000	0.001	0.12	58.75	0.12	58.15
12Y	2.248	2.245	2.356	2.357	0.000	0.003	0.000	−0.001	0.00	58.27	0.00	58.03
15Y	2.214	2.213	2.307	2.305	0.000	0.001	0.000	0.002	0.00	57.31	0.00	56.51
20Y	2.203	2.212	2.271	2.275	0.000	−0.008	0.000	−0.004	0.00	55.40	0.00	54.63
25Y	2.147	2.160	2.192	2.195	0.000	−0.013	0.000	−0.003	0.00	54.32	0.00	54.16
30Y	2.180	2.192	2.219	2.219	0.000	−0.010	0.000	−0.000	0.00	55.16	0.00	55.33

The out-of-sample scaled VAR beats the matched random walk on RMSE at 12 out of 15 tenors, with positive R^2 from 1Y to 15Y, peaking at 0.012 at 5Y–6Y. Directional accuracy is strong throughout, reaching 59.95% at 6Y. The long end (20Y–30Y) shows negative R^2 and marginally higher RMSE than the random walk, indicating that the magnitude calibration is less reliable at very long maturities in the out-of-sample setting.

The hybrid beats its matched random walk only on RMSE at 3 out of 15 tenors. R^2 is mostly small and negative, peaking at 0.002. Directional accuracy is consistently above 50% throughout and exceeds 58% from 9Y to 12Y but does not significantly beat the VAR.

These results partially validate the central claims of the framework. The directional signal is robust for both models: all tenors for both models show directional accuracy well above the random walk. The magnitude claim is more nuanced: the out-of-sample scaled VAR reliably beats the random walk on RMSE across most of the curve, while the out-of-sample scaled hybrid does not show magnitude advantage. The 12.8% and 10.8% of dates excluded due to non-positive alphas confirm that the stability filter is doing meaningful work, and the results on the remaining dates demonstrate that the two-alpha calibration shows to be a valid framework in normal market environments. Tracking the α 's therefore seems highly useful to understand when the model should be used and when the model breaks down. Full residual statistics for the out-of-sample reconstruction are provided in Appendix A6.3.

8 Conclusion

This thesis develops a framework for forecasting the German zero-coupon yield curve. The analysis proceeds in three stages: fitting the yield curve, forecasting the changes in its principal components, and reconstructing yield levels from those forecasts via a calibrated scaling procedure.

Among the two fitting approaches evaluated, PCA4 consistently outperforms both PCA3 and Nelson-Siegel across all maturities and yield curve regimes. The fourth component materially reduces short-end reconstruction errors and provides a more stable basis across the full lag horizon required by the forecasting models. The Nelson-Siegel model, despite its economic interpretability, fails to capture irregular yield curve shapes. We further show that the rolling PCA loading matrix $\mathbf{L}_t$ is stable over the one-month forecast horizon and over lags of up to roughly 200 trading days, validating its use as a locally consistent coordinate system across both the forecasting and reconstruction stages. These findings motivate the choice of PCA4 as the foundation for the forecasting framework.

The VAR analysis reveals two results that are not immediately obvious. First, although the four principal components are orthogonal by construction, their lagged values are dynamically interdependent: the level factor PC1 is a significant predictor in the equations for all other components, confirming that parallel yield curve shifts propagate into slope and curvature dynamics with a delay. A univariate autoregression per component would discard this information, making the VAR the appropriate linear specification. Second, the out-of-sample results show that the VAR achieves meaningful directional accuracy across all components, with PC1 reaching 72% and a positive out-of-sample R^2 for three of the four components.

At the factor level, the LSTM investigation shows that the choice of input representation matters substantially. Feeding the LSTM the same raw lagged factor differences as the VAR produces inferior results on all metrics for all components, indicating that the linear mean-reversion structure identified in the autocorrelation analysis is not readily extracted by the chosen architecture without an appropriate feature encoding. A likely explanation is the severe parameter-to-observation ratio of the LSTM relative to the VAR, which leaves the network under-identified given the available sample. Replacing the raw inputs with the coordinate-consistent factor representation substantially improves performance, particularly for PC2 and PC4, where the LSTM surpasses the VAR on directional accuracy and R^2 at the factor level. The richest variant, which additionally incorporates implied swaption volatility, achieves the only positive out-of-sample R^2 observed for PC4 across any specification. This suggests that nonlinear dynamics are present in the slope and higher-order components that the VAR does not fully capture, while the level factor PC1 remains best described by the linear model.

The central result of this thesis concerns what happens when these factor forecasts are translated back into yield curve levels. Reconstructing the curve directly from the predicted change in factor momentum, as implied by the second-difference relationship underlying the forecasting target, systematically overstates the magnitude of yield changes: an unscaled reconstruction underperforms the random walk on RMSE at every tenor for every model. We address this with a two-parameter scaling procedure that separately calibrates the momentum term and the predicted factor change, estimated by OLS through the origin against realised yield changes. For the VAR and the hybrid model, both scaling coefficients

are positive and statistically significant in-sample, and remain positive in roughly 90% of rolling out-of-sample windows. The resulting out-of-sample scaled VAR reconstruction beats the random walk on RMSE across the short and medium end of the curve (1Y–15Y), with positive out-of-sample R^2 over the same range, and achieves directional accuracy of roughly 54–60% per tenor, peaking around the 5–6 year maturities. This is, to our knowledge, the clearest evidence in this thesis that the factor forecasts carry exploitable information at the one-month horizon once correctly rescaled, even though the framework does not improve on the random walk at the very long end (20Y–30Y), where both RMSE and R^2 deteriorate.

The LSTM-based reconstruction tells a different story and is central to understanding why its factor-level gains for PC2 and PC4 do not materialise at the yield curve level. Both scaling coefficients for the LSTM-VAR-FULL-EIG-IV model are negative in-sample and remain negative in over 92% of rolling windows, meaning that the LSTM’s predicted accelerations point in the direction *opposite* to the realised yield change once projected back through the loading matrix. This is not necessarily a contradiction of the factor-level directional accuracy results: a model can correctly predict the sign of $\Delta f_{k,t+h}$ for individual components while the resulting yield curve reconstruction, which depends on the relative magnitudes and the loading structure across all four components jointly, is dominated by errors elsewhere. In practice this means the LSTM’s improvements for PC2 and PC4 are too small, or too poorly aligned with the other components’ errors, to survive the projection back to yield levels. The hybrid model, which combines VAR predictions for PC1 and PC3 with LSTM predictions for PC2 and PC4, inherits positive and stable alphas similar to the pure VAR, confirming that the instability is attributable specifically to the LSTM’s PC1 and PC3 predictions rather than to PC2 and PC4. Despite this, the hybrid does not show a magnitude advantage over its matched random walk out-of-sample, though it achieves directional accuracy comparable to the VAR and is the strongest model at the long end on direction.

The framework as a whole rests on two stability assumptions. The stale-loadings assumption, that $\mathbf{L}_t$ remains a valid basis over the forecast and lag horizon, is supported by the aggregate evidence of Section 4.6: the distribution of reconstruction errors under stale loadings is broadly similar across all positions from -211 to $+21$ trading days, with only a marginal deterioration in the tail at the forecast horizon. While this evidence is aggregated over the full sample rather than conditioned on specific regimes, the similarity in distributional profile across positions suggests the assumption is reasonable for the purposes of this framework. The stale-alpha assumption is more directly assessed on a rolling basis and proves to be the more informative of the two: it holds in normal market environments but breaks down during the 2022–2023 rate hiking cycle, when both scaling coefficients for the VAR and hybrid turn negative before recovering once the rolling training window has absorbed the new rate regime. Rather than viewing this as a weakness of the framework, we argue it is a useful property: tracking $\hat{\alpha}_1(t)$ and $\hat{\alpha}_2(t)$ in real time provides a direct, interpretable signal of whether the reconstruction is currently reliable. A practitioner observing both coefficients turn negative would have a concrete, model-derived reason to disregard the forecast until the estimates stabilize, rather than relying on the reconstruction blindly.

Taken together, the central finding of this thesis is that a correctly scaled linear factor model, primarily the VAR, produces yield curve forecasts that beat the random walk on both RMSE and directional accuracy across the short and medium end of the curve at a one-month horizon, provided the scaling coefficients are monitored and the forecast is withheld when they are non-positive. The LSTM adds genuine value at the factor level

for the second and fourth components, but in its current form does not improve on the VAR once reconstructed to yield levels, and its instability at the reconstruction stage is itself informative about where the linear model's gains originate. Natural directions for future research include a dedicated gate-level inspection along the lines of Nunes et al. (2026), which would allow the contribution of implied volatility and the coordinate-consistent representation to be isolated directly rather than inferred from the variant progression alone. A second direction is extending the evaluation to longer horizons, where Diebold and Li (2006) document that factor-model outperformance over the random walk becomes more pronounced. Finally, exploring alternative LSTM architectures or regularisation schemes may allow the factor-level advantages for PC2 and PC4 to translate into a stable, positively-scaled yield curve reconstruction, which the current specification does not achieve.

Appendix

A1 Loading Stability Results

Distributional statistics of the reconstruction RMSE (19) at each position $p \in \mathcal{P}$, computed across all 6,209 observations. Positions are expressed in trading days relative to t ; negative positions are lagged inputs, Current is $p = 0$, and +21 is the forecast horizon $t + h$. For Nelson-Siegel, RMSE follows (20) and operates on yield levels rather than changes; absolute magnitudes are therefore not directly comparable across methods.

Table 14: Loading stability RMSE by position — PCA3, 6,209 observations.

Position	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
-211	0.000181	0.000191	0.000010	0.000079	0.000124	0.000213	0.002775
-190	0.000181	0.000191	0.000010	0.000079	0.000124	0.000215	0.002789
-169	0.000181	0.000191	0.000010	0.000079	0.000124	0.000214	0.002825
-148	0.000182	0.000193	0.000011	0.000079	0.000124	0.000215	0.002854
-127	0.000182	0.000195	0.000011	0.000079	0.000125	0.000216	0.002895
-106	0.000182	0.000196	0.000010	0.000079	0.000126	0.000215	0.002932
-85	0.000183	0.000197	0.000010	0.000079	0.000126	0.000216	0.002965
-64	0.000184	0.000200	0.000010	0.000079	0.000126	0.000218	0.003033
-43	0.000183	0.000192	0.000010	0.000079	0.000126	0.000218	0.002769
-22	0.000182	0.000187	0.000012	0.000079	0.000128	0.000217	0.002720
Current ($p = 0$)	0.000182	0.000188	0.000012	0.000080	0.000129	0.000219	0.002735
+21	0.000191	0.000222	0.000012	0.000081	0.000130	0.000225	0.003015

Table 15: Loading stability RMSE by position — PCA4, 6,209 observations.

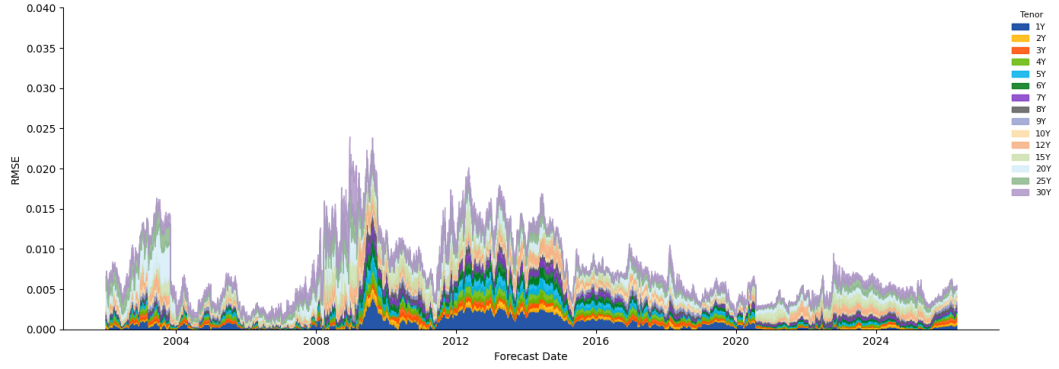
Position	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
-211	0.000129	0.000120	0.000009	0.000060	0.000093	0.000155	0.001530
-190	0.000129	0.000120	0.000010	0.000060	0.000093	0.000155	0.001515
-169	0.000129	0.000121	0.000010	0.000060	0.000092	0.000153	0.001493
-148	0.000128	0.000120	0.000009	0.000059	0.000092	0.000152	0.001466
-127	0.000128	0.000119	0.000009	0.000060	0.000092	0.000151	0.001401
-106	0.000127	0.000116	0.000009	0.000060	0.000091	0.000151	0.001254
-85	0.000127	0.000115	0.000008	0.000060	0.000091	0.000151	0.001213
-64	0.000127	0.000112	0.000008	0.000059	0.000091	0.000151	0.001014
-43	0.000126	0.000108	0.000008	0.000059	0.000091	0.000151	0.001006
-22	0.000125	0.000106	0.000008	0.000060	0.000092	0.000151	0.000971
Current ($p = 0$)	0.000125	0.000106	0.000009	0.000060	0.000093	0.000152	0.001759
+21	0.000134	0.000147	0.000008	0.000061	0.000094	0.000156	0.001927

Table 16: Loading stability RMSE by position — Nelson-Siegel, 6,209 observations.
Operates on yield levels via (20); not directly comparable to PCA figures.

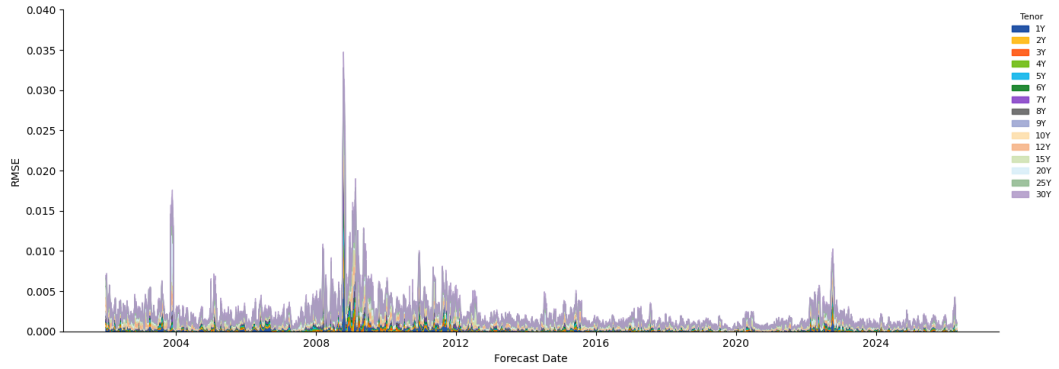
Position	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
−211	0.000625	0.000344	0.000104	0.000363	0.000539	0.000848	0.002239
−190	0.000624	0.000341	0.000105	0.000364	0.000538	0.000847	0.002219
−169	0.000621	0.000339	0.000108	0.000360	0.000536	0.000847	0.002189
−148	0.000619	0.000338	0.000112	0.000358	0.000536	0.000844	0.002153
−127	0.000617	0.000337	0.000112	0.000357	0.000535	0.000840	0.002153
−106	0.000614	0.000336	0.000113	0.000356	0.000533	0.000836	0.002153
−85	0.000612	0.000334	0.000113	0.000352	0.000531	0.000831	0.002153
−64	0.000612	0.000337	0.000113	0.000348	0.000528	0.000829	0.002153
−43	0.000616	0.000343	0.000113	0.000347	0.000528	0.000827	0.002153
−22	0.000621	0.000350	0.000113	0.000351	0.000528	0.000831	0.002319
Current ($p = 0$)	0.000630	0.000361	0.000112	0.000359	0.000530	0.000838	0.002318
+21	0.000640	0.000373	0.000112	0.000364	0.000533	0.000858	0.002317

A2 Reconstruction Error by Tenor

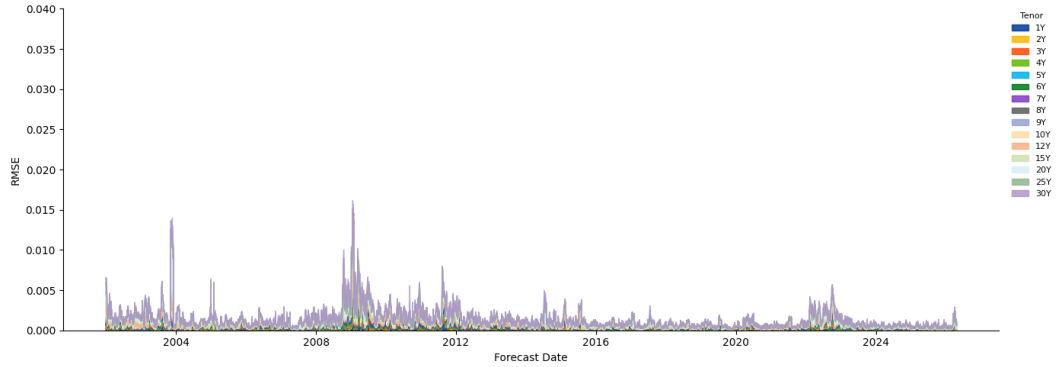
Figure 10 plots reconstruction RMSE per tenor over time for all three methods.



(a) NS



(b) PCA3



(c) PCA4

Figure 10: Reconstruction RMSE per tenor over time for NS, PCA3 and PCA4.

Per-tenor absolute reconstruction errors at the forecast horizon $t+h$, using stale loadings estimated at t . For PCA the reconstruction follows (14); for Nelson-Siegel it follows (7). Errors are computed across all 6,209 observations.

Table 17: Reconstruction errors by tenor — PCA3, 6,209 observations.

Tenor	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
1Y	0.000528	0.000463	0.000000	0.000059	0.000135	0.000285	0.008535
2Y	0.000220	0.000192	0.000000	0.000022	0.000050	0.000120	0.002964
3Y	0.000135	0.000103	0.000000	0.000027	0.000057	0.000107	0.000991
4Y	0.000172	0.000136	0.000000	0.000032	0.000070	0.000134	0.001918
5Y	0.000201	0.000163	0.000000	0.000032	0.000074	0.000147	0.002590
6Y	0.000200	0.000168	0.000000	0.000028	0.000063	0.000134	0.002710
7Y	0.000198	0.000171	0.000000	0.000023	0.000054	0.000113	0.002634
8Y	0.000198	0.000167	0.000000	0.000026	0.000057	0.000121	0.002237
9Y	0.000206	0.000166	0.000000	0.000031	0.000075	0.000146	0.001746
10Y	0.000208	0.000163	0.000000	0.000037	0.000083	0.000157	0.001957
12Y	0.000334	0.000261	0.000000	0.000053	0.000119	0.000255	0.002364
15Y	0.000490	0.000404	0.000000	0.000062	0.000145	0.000326	0.003834
20Y	0.000425	0.000392	0.000000	0.000040	0.000092	0.000181	0.006969
25Y	0.000241	0.000187	0.000000	0.000042	0.000094	0.000186	0.001664
30Y	0.000286	0.000226	0.000000	0.000048	0.000107	0.000209	0.002268

Table 18: Reconstruction errors by tenor — PCA4, 6,209 observations.

Tenor	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
1Y	0.000227	0.000180	0.000000	0.000033	0.000076	0.000168	0.001640
2Y	0.000075	0.000055	0.000000	0.000016	0.000034	0.000065	0.000586
3Y	0.000122	0.000095	0.000000	0.000024	0.000050	0.000093	0.001066
4Y	0.000131	0.000103	0.000000	0.000021	0.000048	0.000101	0.000997
5Y	0.000121	0.000093	0.000000	0.000021	0.000049	0.000099	0.001123
6Y	0.000097	0.000072	0.000000	0.000019	0.000041	0.000084	0.000669
7Y	0.000084	0.000060	0.000000	0.000018	0.000040	0.000078	0.000452
8Y	0.000104	0.000077	0.000000	0.000021	0.000046	0.000091	0.000784
9Y	0.000137	0.000107	0.000000	0.000024	0.000056	0.000109	0.001416
10Y	0.000157	0.000128	0.000000	0.000024	0.000054	0.000112	0.001842
12Y	0.000308	0.000254	0.000000	0.000033	0.000086	0.000209	0.002631
15Y	0.000235	0.000191	0.000000	0.000033	0.000084	0.000172	0.002804
20Y	0.000428	0.000396	0.000000	0.000039	0.000087	0.000175	0.006941
25Y	0.000187	0.000150	0.000000	0.000030	0.000068	0.000129	0.001438
30Y	0.000230	0.000185	0.000000	0.000034	0.000081	0.000165	0.002004

Table 19: Reconstruction errors by tenor — Nelson-Siegel, 6,209 observations. Operates on yield levels via (7); not directly comparable to PCA figures.

Tenor	Mean	Std. Dev.	Min	Q25	Median	Q75	Max
1Y	0.001157	0.000810	0.000000	0.000203	0.000548	0.001170	0.004470
2Y	0.000367	0.000249	0.000000	0.000089	0.000202	0.000374	0.001599
3Y	0.000401	0.000233	0.000000	0.000134	0.000294	0.000481	0.001329
4Y	0.000625	0.000392	0.000000	0.000165	0.000384	0.000700	0.001857
5Y	0.000672	0.000450	0.000000	0.000160	0.000355	0.000701	0.002220
6Y	0.000622	0.000426	0.000000	0.000159	0.000294	0.000635	0.002335
7Y	0.000516	0.000344	0.000000	0.000136	0.000295	0.000530	0.002113
8Y	0.000361	0.000225	0.000000	0.000109	0.000237	0.000395	0.001431
9Y	0.000330	0.000202	0.000000	0.000093	0.000209	0.000402	0.001022
10Y	0.000438	0.000264	0.000000	0.000143	0.000303	0.000507	0.002050
12Y	0.000842	0.000559	0.000000	0.000193	0.000483	0.000869	0.003632
15Y	0.001097	0.000744	0.000000	0.000245	0.000606	0.001083	0.004287
20Y	0.001160	0.000842	0.000000	0.000281	0.000553	0.000980	0.004816
25Y	0.000845	0.000585	0.000000	0.000200	0.000452	0.000768	0.004635
30Y	0.000840	0.000621	0.000000	0.000209	0.000462	0.000715	0.005808

A3 ACF and PACF of Principal Component Factor Changes

ACF and PACF values of the 21-day factor change series $\Delta f_{k,t}$ as defined in (16), evaluated at the lags in $\mathcal{L}^*$ and beyond. The ACF at lag ℓ follows (17); the PACF is obtained via the Yule-Walker equations. Significance is assessed against the 95% bound $\pm 1.96/\sqrt{N}$, denoted $*$.

Table 20: ACF and PACF of $\Delta f_{k,t}$ at selected lags. $*$ denotes significance at the 95% level ($|r| > 1.96/\sqrt{N}$).

ℓ	$\Delta f_{1,t}$		$\Delta f_{2,t}$		$\Delta f_{3,t}$		$\Delta f_{4,t}$	
	ACF	PACF	ACF	PACF	ACF	PACF	ACF	PACF
22	-0.4888*	+0.6106*	-0.5469*	+0.4928*	-0.4420*	+0.4552*	-0.5234*	+0.5856*
43	+0.0259*	+0.4546*	+0.1131*	+0.2529*	-0.0055	+0.1902*	+0.0766*	+0.3442*
64	-0.0532*	+0.3397*	+0.0190	+0.0916*	+0.0867*	+0.0666*	-0.0741*	+0.2313*
85	+0.0994*	+0.2686*	-0.0311*	+0.0083	-0.0277*	+0.0124	+0.1467*	+0.0807*
106	-0.0318*	+0.2107*	-0.0017	+0.0046	+0.0139	-0.0072	-0.1506*	+0.0392*
127	-0.0319*	+0.1374*	+0.0275*	-0.0258*	+0.0375*	-0.0195	+0.0773*	+0.0084
148	-0.0359*	+0.1295*	-0.0759*	+0.0185	-0.1326*	+0.0129	+0.0081	+0.0130
169	+0.0437*	+0.0750*	+0.1163*	+0.0006	+0.1142*	-0.0154	-0.0471*	-0.0606*
190	+0.0551*	+0.0551*	-0.1292*	-0.0123	+0.0032	-0.0150	+0.0193	-0.0116
211	-0.0912*	+0.0299*	+0.0707*	-0.0080	-0.0806*	+0.0020	+0.0256*	+0.0156
232	+0.0176	+0.0218	+0.0295*	-0.0049	-0.0136	+0.0072	-0.0045	+0.0086
253	+0.0402*	+0.0027	-0.0434*	+0.0180	+0.0444*	+0.0065	-0.0889*	-0.0212
274	-0.0076	+0.0061	-0.0016	-0.0089	-0.0158	-0.0107	+0.1333*	-0.0339*
295	+0.0036	-0.0172	-0.0004	+0.0164	-0.0260*	+0.0035	-0.1346*	-0.0240

A4 In-Sample VAR Coefficient Tables

Full coefficient tables for the four equations of the in-sample VAR (21), estimated on $N = 5,998$ observations. Each equation regresses $\Delta f_{k,t}$ on ten lags $\ell \in \mathcal{L}^*$ of all four components. Reported are the OLS coefficient $\hat{\phi}_{k,j}^{(\ell)}$, standard deviation of the estimate, t -statistic, p -value, and 95% confidence interval. Significance: $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

Table 21: In-sample VAR coefficients — $\Delta f_{1,t}$ equation (intercept = +0.000143, $R^2 = 0.4683$, $N = 5,998$).

ℓ	Predictor	$\hat{\phi}_{1,j}^{(\ell)}$	Std. Dev.	t	p	95% CI
22	$\Delta f_{1,t}$	-0.9101***	0.0136	-66.822	0.0000	[-0.937, -0.883]
	$\Delta f_{2,t}$	+0.0733	0.0406	1.807	0.0707	[-0.006, +0.153]
	$\Delta f_{3,t}$	+0.4002***	0.0822	4.869	0.0000	[+0.239, +0.561]
	$\Delta f_{4,t}$	+0.5955***	0.1190	5.002	0.0000	[+0.362, +0.829]
43	$\Delta f_{1,t}$	-0.8148***	0.0189	-43.190	0.0000	[-0.852, -0.778]
	$\Delta f_{2,t}$	+0.1214*	0.0506	2.400	0.0164	[+0.022, +0.221]
	$\Delta f_{3,t}$	+0.3974***	0.0969	4.101	0.0000	[+0.207, +0.587]
	$\Delta f_{4,t}$	+0.4054**	0.1534	2.643	0.0082	[+0.105, +0.706]
64	$\Delta f_{1,t}$	-0.7250***	0.0221	-32.734	0.0000	[-0.768, -0.682]
	$\Delta f_{2,t}$	-0.0325	0.0529	-0.614	0.5395	[-0.136, +0.071]
	$\Delta f_{3,t}$	+0.0060	0.1001	0.060	0.9525	[-0.190, +0.202]
	$\Delta f_{4,t}$	-0.0468	0.1690	-0.277	0.7818	[-0.378, +0.285]
85	$\Delta f_{1,t}$	-0.5201***	0.0238	-21.865	0.0000	[-0.567, -0.474]
	$\Delta f_{2,t}$	-0.0556	0.0532	-1.045	0.2962	[-0.160, +0.049]
	$\Delta f_{3,t}$	-0.7329***	0.1002	-7.317	0.0000	[-0.929, -0.537]
	$\Delta f_{4,t}$	-0.3748*	0.1755	-2.136	0.0327	[-0.719, -0.031]
106	$\Delta f_{1,t}$	-0.4371***	0.0241	-18.150	0.0000	[-0.484, -0.390]
	$\Delta f_{2,t}$	-0.3181***	0.0536	-5.938	0.0000	[-0.423, -0.213]
	$\Delta f_{3,t}$	-0.5556***	0.0995	-5.586	0.0000	[-0.751, -0.361]
	$\Delta f_{4,t}$	-0.0875	0.1766	-0.496	0.6200	[-0.434, +0.259]
127	$\Delta f_{1,t}$	-0.3825***	0.0239	-15.973	0.0000	[-0.429, -0.336]
	$\Delta f_{2,t}$	-0.3753***	0.0534	-7.030	0.0000	[-0.480, -0.271]
	$\Delta f_{3,t}$	-0.9185***	0.0987	-9.303	0.0000	[-1.112, -0.725]
	$\Delta f_{4,t}$	-0.9548***	0.1774	-5.382	0.0000	[-1.303, -0.607]
148	$\Delta f_{1,t}$	-0.3387***	0.0229	-14.763	0.0000	[-0.384, -0.294]
	$\Delta f_{2,t}$	-0.3264***	0.0533	-6.121	0.0000	[-0.431, -0.222]
	$\Delta f_{3,t}$	-0.6631***	0.0988	-6.714	0.0000	[-0.857, -0.470]
	$\Delta f_{4,t}$	-1.2480***	0.1764	-7.074	0.0000	[-1.594, -0.902]
169	$\Delta f_{1,t}$	-0.2070***	0.0211	-9.797	0.0000	[-0.249, -0.166]
	$\Delta f_{2,t}$	-0.4332***	0.0530	-8.171	0.0000	[-0.537, -0.329]
	$\Delta f_{3,t}$	-0.2754**	0.0984	-2.799	0.0051	[-0.468, -0.083]
	$\Delta f_{4,t}$	-1.2359***	0.1692	-7.303	0.0000	[-1.568, -0.904]
190	$\Delta f_{1,t}$	-0.0987***	0.0184	-5.362	0.0000	[-0.135, -0.063]
	$\Delta f_{2,t}$	-0.5259***	0.0504	-10.428	0.0000	[-0.625, -0.427]
	$\Delta f_{3,t}$	+0.2745**	0.0943	2.912	0.0036	[+0.090, +0.459]
	$\Delta f_{4,t}$	-0.9635***	0.1556	-6.193	0.0000	[-1.268, -0.659]
211	$\Delta f_{1,t}$	-0.0923***	0.0135	-6.841	0.0000	[-0.119, -0.066]
	$\Delta f_{2,t}$	-0.2098***	0.0397	-5.278	0.0000	[-0.288, -0.132]
	$\Delta f_{3,t}$	+0.2080*	0.0808	2.574	0.0101	[+0.050, +0.367]
	$\Delta f_{4,t}$	-0.2239	0.1197	-1.870	0.0616	[-0.459, +0.011]

Table 22: In-sample VAR coefficients — $\Delta f_{2,t}$ equation (intercept = +0.000143, $R^2 = 0.4668$, $N = 5,998$).

ℓ	Predictor	$\hat{\phi}_{2,j}^{(\ell)}$	Std. Dev.	t	p	95% CI
22	$\Delta f_{1,t}$	-0.0024	0.0045	-0.527	0.5982	[-0.011, +0.007]
	$\Delta f_{2,t}$	-0.6859***	0.0134	-51.003	0.0000	[-0.712, -0.660]
	$\Delta f_{3,t}$	-0.0182	0.0273	-0.668	0.5042	[-0.072, +0.035]
	$\Delta f_{4,t}$	-0.1430***	0.0395	-3.622	0.0003	[-0.220, -0.066]
43	$\Delta f_{1,t}$	-0.0220***	0.0063	-3.524	0.0004	[-0.034, -0.010]
	$\Delta f_{2,t}$	-0.3733***	0.0168	-22.270	0.0000	[-0.406, -0.341]
	$\Delta f_{3,t}$	-0.1913***	0.0321	-5.954	0.0000	[-0.254, -0.128]
	$\Delta f_{4,t}$	+0.4705***	0.0509	9.251	0.0000	[+0.371, +0.570]
64	$\Delta f_{1,t}$	-0.0474***	0.0073	-6.456	0.0000	[-0.062, -0.033]
	$\Delta f_{2,t}$	-0.1669***	0.0176	-9.507	0.0000	[-0.201, -0.133]
	$\Delta f_{3,t}$	-0.2035***	0.0332	-6.132	0.0000	[-0.269, -0.139]
	$\Delta f_{4,t}$	-0.0042	0.0560	-0.076	0.9397	[-0.114, +0.106]
85	$\Delta f_{1,t}$	-0.0330***	0.0079	-4.179	0.0000	[-0.048, -0.018]
	$\Delta f_{2,t}$	-0.1419***	0.0176	-8.047	0.0000	[-0.177, -0.107]
	$\Delta f_{3,t}$	+0.0391	0.0332	1.176	0.2396	[-0.026, +0.104]
	$\Delta f_{4,t}$	+0.0389	0.0582	0.669	0.5035	[-0.075, +0.153]
106	$\Delta f_{1,t}$	-0.0463***	0.0080	-5.800	0.0000	[-0.062, -0.031]
	$\Delta f_{2,t}$	-0.0651***	0.0178	-3.665	0.0002	[-0.100, -0.030]
	$\Delta f_{3,t}$	+0.1620***	0.0330	4.912	0.0000	[+0.097, +0.227]
	$\Delta f_{4,t}$	+0.2107***	0.0585	3.599	0.0003	[+0.096, +0.326]
127	$\Delta f_{1,t}$	-0.0341***	0.0079	-4.293	0.0000	[-0.050, -0.019]
	$\Delta f_{2,t}$	-0.0429*	0.0177	-2.426	0.0153	[-0.078, -0.008]
	$\Delta f_{3,t}$	+0.1059**	0.0327	3.235	0.0012	[+0.042, +0.170]
	$\Delta f_{4,t}$	+0.2690***	0.0588	4.573	0.0000	[+0.154, +0.384]
148	$\Delta f_{1,t}$	-0.0192*	0.0076	-2.525	0.0116	[-0.034, -0.004]
	$\Delta f_{2,t}$	-0.0913***	0.0177	-5.161	0.0000	[-0.126, -0.057]
	$\Delta f_{3,t}$	+0.0741*	0.0327	2.263	0.0237	[+0.010, +0.138]
	$\Delta f_{4,t}$	+0.2168***	0.0585	3.707	0.0002	[+0.102, +0.332]
169	$\Delta f_{1,t}$	-0.0272***	0.0070	-3.883	0.0001	[-0.041, -0.014]
	$\Delta f_{2,t}$	-0.0165	0.0176	-0.938	0.3482	[-0.051, +0.018]
	$\Delta f_{3,t}$	+0.0813*	0.0326	2.493	0.0127	[+0.017, +0.145]
	$\Delta f_{4,t}$	+0.5356***	0.0561	9.546	0.0000	[+0.426, +0.646]
190	$\Delta f_{1,t}$	-0.0278***	0.0061	-4.557	0.0000	[-0.040, -0.016]
	$\Delta f_{2,t}$	-0.0675***	0.0167	-4.037	0.0001	[-0.100, -0.035]
	$\Delta f_{3,t}$	+0.1058***	0.0313	3.384	0.0007	[+0.045, +0.167]
	$\Delta f_{4,t}$	+0.1752***	0.0516	3.396	0.0007	[+0.074, +0.276]
211	$\Delta f_{1,t}$	-0.0069	0.0045	-1.550	0.1212	[-0.016, +0.002]
	$\Delta f_{2,t}$	-0.0242	0.0132	-1.836	0.0664	[-0.050, +0.002]
	$\Delta f_{3,t}$	-0.1025***	0.0268	-3.826	0.0001	[-0.155, -0.050]
	$\Delta f_{4,t}$	+0.2660***	0.0397	6.701	0.0000	[+0.188, +0.344]

Table 23: In-sample VAR coefficients — $\Delta f_{3,t}$ equation (intercept = -0.000059 , $R^2 = 0.3362$, $N = 5,998$).

ℓ	Predictor	$\hat{\phi}_{3,j}^{(\ell)}$	Std. Dev.	t	p	95% CI
22	$\Delta f_{1,t}$	+0.0150***	0.0022	6.672	0.0000	[+0.011, +0.019]
	$\Delta f_{2,t}$	+0.0123	0.0067	1.833	0.0669	[−0.001, +0.025]
	$\Delta f_{3,t}$	−0.5781***	0.0135	−42.668	0.0000	[−0.605, −0.552]
	$\Delta f_{4,t}$	−0.0227	0.0196	−1.154	0.2484	[−0.061, +0.016]
43	$\Delta f_{1,t}$	+0.0325***	0.0031	10.453	0.0000	[+0.026, +0.039]
	$\Delta f_{2,t}$	+0.0046	0.0083	0.547	0.5845	[−0.012, +0.021]
	$\Delta f_{3,t}$	−0.2933***	0.0160	−18.367	0.0000	[−0.325, −0.262]
	$\Delta f_{4,t}$	−0.0652*	0.0253	−2.577	0.0100	[−0.115, −0.016]
64	$\Delta f_{1,t}$	+0.0259***	0.0037	7.082	0.0000	[+0.019, +0.033]
	$\Delta f_{2,t}$	+0.0179*	0.0087	2.047	0.0407	[+0.001, +0.035]
	$\Delta f_{3,t}$	−0.0777***	0.0165	−4.710	0.0000	[−0.110, −0.045]
	$\Delta f_{4,t}$	−0.1037***	0.0279	−3.723	0.0002	[−0.158, −0.049]
85	$\Delta f_{1,t}$	+0.0408***	0.0039	10.414	0.0000	[+0.033, +0.049]
	$\Delta f_{2,t}$	+0.0320***	0.0088	3.646	0.0003	[+0.015, +0.049]
	$\Delta f_{3,t}$	+0.0263	0.0165	1.590	0.1119	[−0.006, +0.059]
	$\Delta f_{4,t}$	−0.0676*	0.0289	−2.335	0.0196	[−0.124, −0.011]
106	$\Delta f_{1,t}$	+0.0312***	0.0040	7.861	0.0000	[+0.023, +0.039]
	$\Delta f_{2,t}$	−0.0025	0.0088	−0.278	0.7810	[−0.020, +0.015]
	$\Delta f_{3,t}$	+0.0937***	0.0164	5.717	0.0000	[+0.062, +0.126]
	$\Delta f_{4,t}$	+0.2235***	0.0291	7.678	0.0000	[+0.166, +0.281]
127	$\Delta f_{1,t}$	+0.0064	0.0039	1.632	0.1027	[−0.001, +0.014]
	$\Delta f_{2,t}$	+0.0024	0.0088	0.271	0.7862	[−0.015, +0.020]
	$\Delta f_{3,t}$	+0.0662***	0.0163	4.067	0.0000	[+0.034, +0.098]
	$\Delta f_{4,t}$	+0.2066***	0.0292	7.063	0.0000	[+0.149, +0.264]
148	$\Delta f_{1,t}$	+0.0135***	0.0038	3.559	0.0004	[+0.006, +0.021]
	$\Delta f_{2,t}$	+0.0145	0.0088	1.644	0.1002	[−0.003, +0.032]
	$\Delta f_{3,t}$	−0.0664***	0.0163	−4.077	0.0000	[−0.098, −0.035]
	$\Delta f_{4,t}$	+0.1030***	0.0291	3.542	0.0004	[+0.046, +0.160]
169	$\Delta f_{1,t}$	+0.0144***	0.0035	4.126	0.0000	[+0.008, +0.021]
	$\Delta f_{2,t}$	−0.0273**	0.0087	−3.123	0.0018	[−0.044, −0.010]
	$\Delta f_{3,t}$	+0.0140	0.0162	0.863	0.3882	[−0.018, +0.046]
	$\Delta f_{4,t}$	+0.0658*	0.0279	2.358	0.0184	[+0.011, +0.121]
190	$\Delta f_{1,t}$	+0.0083**	0.0030	2.741	0.0061	[+0.002, +0.014]
	$\Delta f_{2,t}$	−0.0190*	0.0083	−2.285	0.0223	[−0.035, −0.003]
	$\Delta f_{3,t}$	+0.0137	0.0155	0.881	0.3786	[−0.017, +0.044]
	$\Delta f_{4,t}$	−0.0533*	0.0256	−2.079	0.0377	[−0.104, −0.003]
211	$\Delta f_{1,t}$	+0.0031	0.0022	1.379	0.1680	[−0.001, +0.007]
	$\Delta f_{2,t}$	+0.0024	0.0066	0.362	0.7176	[−0.011, +0.015]
	$\Delta f_{3,t}$	−0.0363**	0.0133	−2.726	0.0064	[−0.062, −0.010]
	$\Delta f_{4,t}$	+0.0085	0.0197	0.428	0.6685	[−0.030, +0.047]

Table 24: In-sample VAR coefficients — $\Delta f_{4,t}$ equation (intercept = -0.000000 , $R^2 = 0.4581$, $N = 5,998$).

ℓ	Predictor	$\hat{\phi}_{4,j}^{(\ell)}$	Std. Dev.	t	p	95% CI
22	$\Delta f_{1,t}$	+0.0102***	0.0016	6.524	0.0000	[+0.007, +0.013]
	$\Delta f_{2,t}$	-0.0195***	0.0047	-4.170	0.0000	[-0.029, -0.010]
	$\Delta f_{3,t}$	+0.0090	0.0095	0.954	0.3401	[-0.010, +0.028]
	$\Delta f_{4,t}$	-0.7699***	0.0137	-56.179	0.0000	[-0.797, -0.743]
43	$\Delta f_{1,t}$	+0.0168***	0.0022	7.742	0.0000	[+0.013, +0.021]
	$\Delta f_{2,t}$	-0.0136*	0.0058	-2.333	0.0197	[-0.025, -0.002]
	$\Delta f_{3,t}$	+0.0078	0.0112	0.695	0.4871	[-0.014, +0.030]
	$\Delta f_{4,t}$	-0.5772***	0.0177	-32.692	0.0000	[-0.612, -0.543]
64	$\Delta f_{1,t}$	+0.0182***	0.0025	7.142	0.0000	[+0.013, +0.023]
	$\Delta f_{2,t}$	-0.0128*	0.0061	-2.099	0.0359	[-0.025, -0.001]
	$\Delta f_{3,t}$	+0.0773***	0.0115	6.711	0.0000	[+0.055, +0.100]
	$\Delta f_{4,t}$	-0.4002***	0.0195	-20.574	0.0000	[-0.438, -0.362]
85	$\Delta f_{1,t}$	+0.0153***	0.0027	5.586	0.0000	[+0.010, +0.021]
	$\Delta f_{2,t}$	-0.0376***	0.0061	-6.144	0.0000	[-0.050, -0.026]
	$\Delta f_{3,t}$	-0.0384***	0.0115	-3.330	0.0009	[-0.061, -0.016]
	$\Delta f_{4,t}$	-0.2254***	0.0202	-11.161	0.0000	[-0.265, -0.186]
106	$\Delta f_{1,t}$	+0.0116***	0.0028	4.186	0.0000	[+0.006, +0.017]
	$\Delta f_{2,t}$	+0.0060	0.0062	0.976	0.3291	[-0.006, +0.018]
	$\Delta f_{3,t}$	-0.0202	0.0114	-1.761	0.0783	[-0.043, +0.002]
	$\Delta f_{4,t}$	-0.1961***	0.0203	-9.648	0.0000	[-0.236, -0.156]
127	$\Delta f_{1,t}$	+0.0120***	0.0028	4.342	0.0000	[+0.007, +0.017]
	$\Delta f_{2,t}$	-0.0396***	0.0061	-6.445	0.0000	[-0.052, -0.028]
	$\Delta f_{3,t}$	-0.0554***	0.0114	-4.874	0.0000	[-0.078, -0.033]
	$\Delta f_{4,t}$	-0.1328***	0.0204	-6.503	0.0000	[-0.173, -0.093]
148	$\Delta f_{1,t}$	+0.0154***	0.0026	5.829	0.0000	[+0.010, +0.021]
	$\Delta f_{2,t}$	-0.0408***	0.0061	-6.645	0.0000	[-0.053, -0.029]
	$\Delta f_{3,t}$	-0.0001	0.0114	-0.013	0.9896	[-0.022, +0.022]
	$\Delta f_{4,t}$	-0.1198***	0.0203	-5.899	0.0000	[-0.160, -0.080]
169	$\Delta f_{1,t}$	+0.0173***	0.0024	7.113	0.0000	[+0.013, +0.022]
	$\Delta f_{2,t}$	-0.0133*	0.0061	-2.184	0.0290	[-0.025, -0.001]
	$\Delta f_{3,t}$	-0.0464***	0.0113	-4.095	0.0000	[-0.069, -0.024]
	$\Delta f_{4,t}$	-0.1029***	0.0195	-5.284	0.0000	[-0.141, -0.065]
190	$\Delta f_{1,t}$	+0.0120***	0.0021	5.668	0.0000	[+0.008, +0.016]
	$\Delta f_{2,t}$	+0.0123*	0.0058	2.127	0.0335	[+0.001, +0.024]
	$\Delta f_{3,t}$	+0.0373***	0.0109	3.433	0.0006	[+0.016, +0.059]
	$\Delta f_{4,t}$	+0.0126	0.0179	0.702	0.4829	[-0.023, +0.048]
211	$\Delta f_{1,t}$	+0.0006	0.0016	0.360	0.7185	[-0.003, +0.004]
	$\Delta f_{2,t}$	-0.0042	0.0046	-0.922	0.3567	[-0.013, +0.005]
	$\Delta f_{3,t}$	+0.0213*	0.0093	2.289	0.0221	[+0.003, +0.040]
	$\Delta f_{4,t}$	+0.0260	0.0138	1.889	0.0589	[-0.001, +0.053]

A5 Out-of-Sample LSTM Forecast Results

Performance and residual statistics for all LSTM variants at the factor level, evaluated over $N = 2,977$ out-of-sample predictions (2014-07-24 to 2026-03-13). Metrics follow the same definitions as Section 5.2: directional accuracy (27), $R^2 = 1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$, RMSE, and AIC/BIC from the Gaussian log-likelihood with 40 parameters. Residuals are defined as $e_{k,t} = \Delta f_{k,t} - \widehat{\Delta f}_{k,t}$.

Table 25: Out-of-sample rolling LSTM performance — LSTM-VAR, $\tau = 3,000$, $N = 2,977$.

$\Delta f_{k,t}$	N	Accuracy	R^2	RMSE	AIC	BIC
$\Delta f_{1,t}$	2977	64.70%	0.0949	0.010304	-18710.59	-18464.64
$\Delta f_{2,t}$	2977	56.73%	-0.2481	0.003134	-25796.53	-25550.58
$\Delta f_{3,t}$	2977	59.05%	-0.1547	0.001291	-31075.71	-30829.77
$\Delta f_{4,t}$	2977	57.10%	-0.6572	0.000665	-35030.10	-34784.16

Table 26: Out-of-sample forecast residual statistics — LSTM-VAR.

$\Delta f_{k,t}$	Mean	Std. Dev.	Min	Q25	Q75	Max
$\Delta f_{1,t}$	0.000495	0.010292	-0.047743	-0.005222	0.006387	0.053006
$\Delta f_{2,t}$	0.000019	0.003134	-0.012220	-0.001668	0.001606	0.015873
$\Delta f_{3,t}$	0.000073	0.001289	-0.004903	-0.000682	0.000792	0.009416
$\Delta f_{4,t}$	-0.000018	0.000664	-0.002706	-0.000385	0.000314	0.005576

Table 27: Out-of-sample rolling LSTM performance — LSTM-FULL, $\tau = 3,000$, $N = 2,977$.

$\Delta f_{k,t}$	N	Accuracy	R^2	RMSE	AIC	BIC
$\Delta f_{1,t}$	2977	70.27%	0.2355	0.009470	-19213.25	-18967.30
$\Delta f_{2,t}$	2977	68.26%	0.1577	0.002575	-26967.33	-26721.39
$\Delta f_{3,t}$	2977	65.97%	0.0955	0.001143	-31802.90	-31556.95
$\Delta f_{4,t}$	2977	65.50%	-0.1289	0.000549	-36172.71	-35926.77

Table 28: Out-of-sample forecast residual statistics — LSTM-FULL.

$\Delta f_{k,t}$	Mean	Std. Dev.	Min	Q25	Q75	Max
$\Delta f_{1,t}$	0.000686	0.009445	-0.042103	-0.004543	0.005792	0.049596
$\Delta f_{2,t}$	-0.000190	0.002568	-0.011084	-0.001502	0.001169	0.010648
$\Delta f_{3,t}$	0.000047	0.001142	-0.004319	-0.000612	0.000653	0.006796
$\Delta f_{4,t}$	-0.000003	0.000549	-0.002388	-0.000280	0.000269	0.003276

Table 29: Out-of-sample rolling LSTM performance — LSTM-FULL-EIG, $\tau = 3,000$, $N = 2,977$.

$\Delta f_{k,t}$	N	Accuracy	R^2	RMSE	AIC	BIC
$\Delta f_{1,t}$	2977	69.70%	0.2569	0.009336	-19297.61	-19051.67
$\Delta f_{2,t}$	2977	65.57%	0.1374	0.002606	-26896.28	-26650.34
$\Delta f_{3,t}$	2977	65.77%	0.1759	0.001091	-32079.85	-31833.91
$\Delta f_{4,t}$	2977	55.59%	-0.1117	0.000544	-36218.65	-35972.71

Table 30: Out-of-sample forecast residual statistics — LSTM-FULL-EIG.

$\Delta f_{k,t}$	Mean	Std. Dev.	Min	Q25	Q75	Max
$\Delta f_{1,t}$	0.000552	0.009320	-0.042103	-0.004527	0.005034	0.051141
$\Delta f_{2,t}$	-0.000048	0.002605	-0.015010	-0.001317	0.001230	0.014038
$\Delta f_{3,t}$	0.000011	0.001091	-0.004824	-0.000599	0.000581	0.006190
$\Delta f_{4,t}$	0.000027	0.000544	-0.002232	-0.000255	0.000296	0.003725

Table 31: Out-of-sample rolling LSTM performance — LSTM-VAR-FULL-EIG, $\tau = 3,000$, $N = 2,977$.

$\Delta f_{k,t}$	N	Accuracy	R^2	RMSE	AIC	BIC
$\Delta f_{1,t}$	2977	71.35%	0.2949	0.009094	-19454.03	-19208.08
$\Delta f_{2,t}$	2977	66.78%	0.1782	0.002543	-27040.68	-26794.73
$\Delta f_{3,t}$	2977	66.31%	0.1922	0.001080	-32139.44	-31893.50
$\Delta f_{4,t}$	2977	58.92%	-0.1155	0.000545	-36208.32	-35962.37

Table 32: Out-of-sample forecast residual statistics — LSTM-VAR-FULL-EIG.

$\Delta f_{k,t}$	Mean	Std. Dev.	Min	Q25	Q75	Max
$\Delta f_{1,t}$	0.000684	0.009069	-0.037416	-0.004265	0.005318	0.046883
$\Delta f_{2,t}$	0.000017	0.002543	-0.013292	-0.001238	0.001255	0.010943
$\Delta f_{3,t}$	-0.000015	0.001080	-0.004451	-0.000632	0.000605	0.006035
$\Delta f_{4,t}$	0.000020	0.000545	-0.003003	-0.000247	0.000268	0.004373

Table 33: Out-of-sample rolling LSTM performance — LSTM-VAR-FULL-EIG-IV, $\tau = 3,000$, $N = 2,977$.

$\Delta f_{k,t}$	N	Accuracy	R^2	RMSE	AIC	BIC
$\Delta f_{1,t}$	2977	71.08%	0.3178	0.008946	-19552.15	-19306.21
$\Delta f_{2,t}$	2977	67.48%	0.2252	0.002469	-27216.10	-26970.15
$\Delta f_{3,t}$	2977	65.40%	0.1980	0.001076	-32160.95	-31915.01
$\Delta f_{4,t}$	2977	60.36%	0.0026	0.000516	-36541.64	-36295.70

Table 34: Out-of-sample forecast residual statistics — LSTM-VAR-FULL-EIG-IV.

$\Delta f_{k,t}$	Mean	Std. Dev.	Min	Q25	Q75	Max
$\Delta f_{1,t}$	0.000724	0.008916	-0.039789	-0.004009	0.005224	0.048186
$\Delta f_{2,t}$	-0.000008	0.002469	-0.012722	-0.001214	0.001181	0.012318
$\Delta f_{3,t}$	0.000007	0.001076	-0.004368	-0.000609	0.000595	0.007374
$\Delta f_{4,t}$	0.000013	0.000516	-0.002072	-0.000263	0.000268	0.003854

A6 Yield Curve Reconstruction Residual Statistics

For each tenor $j \in \{1Y, \dots, 30Y\}$ and observation i , the forecast residual is defined as:

$$e_{i,j} = \hat{y}_{i,j} - y_{i,j}$$

where $\hat{y}_{i,j}$ is the reconstructed yield and $y_{i,j}$ the realised yield. The tables below report the sample mean, standard deviation, minimum, Q25, Q75, and maximum of $e_{i,j}$ across all N observations.

A6.1 Unscaled Reconstruction ($\alpha_1 = \alpha_2 = 1$)

Residuals from the unscaled reconstruction (44) applied to the full out-of-sample window ($N = 2,977$). No magnitude calibration is applied; the results correspond to the directional analysis of Section 7.2.

Table 35: Unscaled reconstruction residuals — VAR, $N = 2,977$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000116	0.001809	-0.011132	-0.000860	0.000601	0.009291
2Y	-0.000085	0.001897	-0.010477	-0.000890	0.000677	0.010472
3Y	-0.000065	0.001997	-0.010439	-0.000929	0.000817	0.011232
4Y	-0.000051	0.002077	-0.011335	-0.001035	0.000936	0.011233
5Y	-0.000044	0.002131	-0.011522	-0.001113	0.001003	0.010793
6Y	-0.000040	0.002169	-0.011266	-0.001192	0.001100	0.010248
7Y	-0.000040	0.002207	-0.010787	-0.001246	0.001154	0.009826
8Y	-0.000041	0.002254	-0.010392	-0.001297	0.001233	0.009866
9Y	-0.000041	0.002308	-0.010247	-0.001343	0.001308	0.010002
10Y	-0.000038	0.002343	-0.009866	-0.001333	0.001338	0.010187
12Y	-0.000029	0.002465	-0.009569	-0.001432	0.001397	0.010716
15Y	-0.000022	0.002507	-0.009816	-0.001509	0.001476	0.010115
20Y	-0.000005	0.002612	-0.010777	-0.001575	0.001515	0.010710
25Y	0.000005	0.002637	-0.011098	-0.001628	0.001533	0.010650
30Y	0.000002	0.002637	-0.011266	-0.001608	0.001541	0.010399

Table 36: Unscaled reconstruction residuals — LSTM-VAR-FULL-EIG-IV, $N = 2,977$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000126	0.001965	-0.014745	-0.000783	0.000613	0.010690
2Y	-0.000145	0.002090	-0.015327	-0.000855	0.000664	0.012215
3Y	-0.000161	0.002205	-0.015487	-0.000991	0.000765	0.012762
4Y	-0.000174	0.002290	-0.015397	-0.001132	0.000892	0.012613
5Y	-0.000184	0.002344	-0.015122	-0.001218	0.000976	0.012140
6Y	-0.000191	0.002374	-0.014587	-0.001303	0.001012	0.011558
7Y	-0.000199	0.002395	-0.013868	-0.001355	0.001051	0.010983
8Y	-0.000206	0.002422	-0.013124	-0.001443	0.001110	0.010467
9Y	-0.000210	0.002453	-0.012429	-0.001482	0.001153	0.010357
10Y	-0.000208	0.002474	-0.012047	-0.001519	0.001173	0.010558
12Y	-0.000203	0.002544	-0.012350	-0.001578	0.001201	0.011203
15Y	-0.000197	0.002575	-0.012468	-0.001601	0.001278	0.010078
20Y	-0.000190	0.002640	-0.013431	-0.001713	0.001340	0.010533
25Y	-0.000188	0.002638	-0.013674	-0.001738	0.001411	0.010156
30Y	-0.000192	0.002639	-0.013793	-0.001732	0.001396	0.010197

Table 37: Unscaled reconstruction residuals — Hybrid, $N = 2,977$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000080	0.001827	-0.011580	-0.000786	0.000598	0.008851
2Y	-0.000062	0.001947	-0.011101	-0.000809	0.000671	0.010613
3Y	-0.000053	0.002058	-0.010646	-0.000932	0.000789	0.011281
4Y	-0.000048	0.002140	-0.010772	-0.001056	0.000926	0.011322
5Y	-0.000046	0.002187	-0.010936	-0.001153	0.001035	0.011007
6Y	-0.000045	0.002212	-0.010778	-0.001213	0.001115	0.010527
7Y	-0.000047	0.002231	-0.010566	-0.001247	0.001186	0.010009
8Y	-0.000050	0.002257	-0.010383	-0.001310	0.001252	0.009894
9Y	-0.000050	0.002287	-0.010258	-0.001346	0.001296	0.009957
10Y	-0.000046	0.002306	-0.009918	-0.001358	0.001327	0.010030
12Y	-0.000035	0.002389	-0.009741	-0.001424	0.001354	0.010236
15Y	-0.000034	0.002429	-0.010003	-0.001480	0.001439	0.009560
20Y	-0.000029	0.002514	-0.011007	-0.001545	0.001480	0.010081
25Y	-0.000024	0.002520	-0.011336	-0.001634	0.001503	0.009892
30Y	-0.000027	0.002521	-0.011485	-0.001627	0.001505	0.009400

Table 38: Unscaled reconstruction residuals — Random Walk, $N = 2,977$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000171	0.001696	-0.010834	-0.000539	0.000531	0.006841
2Y	-0.000173	0.001765	-0.010308	-0.000597	0.000591	0.007507
3Y	-0.000173	0.001836	-0.009891	-0.000720	0.000679	0.007764
4Y	-0.000170	0.001887	-0.009708	-0.000880	0.000778	0.007921
5Y	-0.000165	0.001920	-0.009445	-0.001002	0.000862	0.007741
6Y	-0.000159	0.001943	-0.009059	-0.001082	0.000932	0.007739
7Y	-0.000152	0.001963	-0.008708	-0.001154	0.000997	0.007696
8Y	-0.000145	0.001987	-0.008535	-0.001202	0.001060	0.007692
9Y	-0.000138	0.002011	-0.008405	-0.001220	0.001105	0.007700
10Y	-0.000131	0.002026	-0.008263	-0.001228	0.001141	0.007698
12Y	-0.000120	0.002087	-0.008002	-0.001258	0.001196	0.007845
15Y	-0.000119	0.002132	-0.008155	-0.001291	0.001251	0.007082
20Y	-0.000109	0.002194	-0.008748	-0.001308	0.001248	0.007459
25Y	-0.000103	0.002175	-0.008421	-0.001340	0.001254	0.006821
30Y	-0.000105	0.002176	-0.008609	-0.001344	0.001253	0.006625

A6.2 In-Sample Scaled Reconstruction

Residuals from the two-alpha scaled reconstruction (46) with $\hat{\alpha}_1$ and $\hat{\alpha}_2$ estimated in-sample over the full window ($N = 2,977$). The random walk residuals are identical to Section A6.1 and are not repeated.

Table 39: In-sample scaled reconstruction residuals — VAR, $N = 2,977$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000176	0.001680	-0.011197	-0.000558	0.000507	0.006413
2Y	-0.000175	0.001743	-0.010701	-0.000602	0.000570	0.006870
3Y	-0.000172	0.001811	-0.010299	-0.000737	0.000674	0.007440
4Y	-0.000168	0.001861	-0.009926	-0.000898	0.000762	0.007526
5Y	-0.000162	0.001896	-0.009578	-0.000988	0.000839	0.007379
6Y	-0.000155	0.001921	-0.009137	-0.001072	0.000911	0.007463
7Y	-0.000148	0.001945	-0.008651	-0.001116	0.000971	0.007434
8Y	-0.000141	0.001973	-0.008486	-0.001150	0.001034	0.007448
9Y	-0.000134	0.002000	-0.008359	-0.001191	0.001134	0.007474
10Y	-0.000128	0.002019	-0.008230	-0.001205	0.001179	0.007489
12Y	-0.000117	0.002087	-0.007977	-0.001242	0.001182	0.007626
15Y	-0.000114	0.002130	-0.008038	-0.001262	0.001233	0.006893
20Y	-0.000103	0.002198	-0.008780	-0.001288	0.001242	0.007285
25Y	-0.000096	0.002178	-0.008679	-0.001318	0.001242	0.006931
30Y	-0.000098	0.002177	-0.008868	-0.001322	0.001246	0.006612

Table 40: In-sample scaled reconstruction residuals — LSTM-VAR-FULL-EIG-IV, $N = 2,977$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000182	0.001715	-0.011018	-0.000529	0.000529	0.006293
2Y	-0.000184	0.001768	-0.010503	-0.000589	0.000568	0.006971
3Y	-0.000183	0.001830	-0.010089	-0.000715	0.000674	0.007104
4Y	-0.000180	0.001875	-0.009768	-0.000895	0.000775	0.007210
5Y	-0.000175	0.001906	-0.009497	-0.000999	0.000850	0.007355
6Y	-0.000168	0.001929	-0.009109	-0.001080	0.000925	0.007372
7Y	-0.000160	0.001951	-0.008730	-0.001155	0.000986	0.007294
8Y	-0.000151	0.001975	-0.008403	-0.001196	0.001037	0.007289
9Y	-0.000143	0.002001	-0.008270	-0.001214	0.001103	0.007309
10Y	-0.000137	0.002018	-0.008068	-0.001212	0.001124	0.007334
12Y	-0.000125	0.002081	-0.007949	-0.001249	0.001177	0.007477
15Y	-0.000123	0.002129	-0.008250	-0.001283	0.001240	0.006675
20Y	-0.000113	0.002194	-0.008874	-0.001309	0.001215	0.007080
25Y	-0.000107	0.002173	-0.008346	-0.001325	0.001234	0.006722
30Y	-0.000109	0.002172	-0.008437	-0.001330	0.001229	0.006536

Table 41: In-sample scaled reconstruction residuals — Hybrid, $N = 2,977$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000171	0.001685	-0.011256	-0.000549	0.000512	0.006325
2Y	-0.000171	0.001752	-0.010783	-0.000602	0.000570	0.006867
3Y	-0.000169	0.001822	-0.010384	-0.000747	0.000669	0.007392
4Y	-0.000166	0.001872	-0.010003	-0.000892	0.000760	0.007519
5Y	-0.000161	0.001905	-0.009641	-0.001012	0.000839	0.007429
6Y	-0.000155	0.001929	-0.009180	-0.001072	0.000909	0.007496
7Y	-0.000149	0.001950	-0.008691	-0.001115	0.000978	0.007466
8Y	-0.000142	0.001975	-0.008498	-0.001161	0.001041	0.007474
9Y	-0.000135	0.002000	-0.008366	-0.001195	0.001129	0.007491
10Y	-0.000128	0.002016	-0.008231	-0.001201	0.001169	0.007491
12Y	-0.000117	0.002079	-0.007965	-0.001242	0.001183	0.007612
15Y	-0.000115	0.002123	-0.008035	-0.001262	0.001225	0.006928
20Y	-0.000106	0.002187	-0.008879	-0.001283	0.001223	0.007326
25Y	-0.000100	0.002165	-0.008705	-0.001323	0.001236	0.006903
30Y	-0.000102	0.002163	-0.008896	-0.001329	0.001244	0.006707

A6.3 Out-of-Sample Reconstruction

Residuals from the rolling out-of-sample procedure of Section 7.5, where $[\hat{\alpha}_1(t), \hat{\alpha}_2(t)]$ are estimated on $[t - 2000 - 21, t - 21)$ at each step and only dates with both alphas positive produce a forecast. VAR: $N = 834$, Hybrid: $N = 853$, matched random walks evaluated on the same dates.

Table 42: Out-of-sample reconstruction residuals — VAR, $N = 834$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000033	0.002185	-0.008696	-0.001078	0.001159	0.006803
2Y	-0.000030	0.002311	-0.008410	-0.001248	0.001105	0.007336
3Y	-0.000036	0.002376	-0.008753	-0.001329	0.001092	0.007987
4Y	-0.000049	0.002388	-0.008957	-0.001403	0.001146	0.008249
5Y	-0.000063	0.002360	-0.008992	-0.001399	0.001142	0.008244
6Y	-0.000075	0.002328	-0.008879	-0.001395	0.001138	0.008317
7Y	-0.000087	0.002300	-0.008694	-0.001361	0.001068	0.008271
8Y	-0.000098	0.002284	-0.008525	-0.001326	0.001059	0.008269
9Y	-0.000108	0.002275	-0.008390	-0.001308	0.001073	0.008288
10Y	-0.000113	0.002254	-0.008254	-0.001318	0.001025	0.008309
12Y	-0.000131	0.002241	-0.007994	-0.001334	0.000963	0.008398
15Y	-0.000152	0.002208	-0.007595	-0.001344	0.000949	0.007661
20Y	-0.000184	0.002204	-0.006773	-0.001420	0.000880	0.008048
25Y	-0.000224	0.002149	-0.006143	-0.001470	0.000876	0.007017
30Y	-0.000230	0.002179	-0.006414	-0.001454	0.000834	0.006710

Table 43: Out-of-sample reconstruction residuals — VAR random walk, $N = 834$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000012	0.002190	-0.008397	-0.001044	0.001191	0.006841
2Y	-0.000025	0.002318	-0.008418	-0.001182	0.001090	0.007507
3Y	-0.000044	0.002386	-0.008763	-0.001333	0.001161	0.007457
4Y	-0.000067	0.002399	-0.009011	-0.001389	0.001148	0.007558
5Y	-0.000089	0.002372	-0.008994	-0.001431	0.001175	0.007672
6Y	-0.000108	0.002341	-0.008887	-0.001436	0.001159	0.007739
7Y	-0.000126	0.002312	-0.008708	-0.001414	0.001116	0.007696
8Y	-0.000142	0.002294	-0.008535	-0.001406	0.001083	0.007692
9Y	-0.000156	0.002282	-0.008405	-0.001434	0.001052	0.007700
10Y	-0.000164	0.002259	-0.008263	-0.001460	0.001003	0.007698
12Y	-0.000185	0.002241	-0.008002	-0.001439	0.000914	0.007845
15Y	-0.000214	0.002204	-0.007638	-0.001413	0.000884	0.007082
20Y	-0.000251	0.002189	-0.006823	-0.001476	0.000853	0.007459
25Y	-0.000296	0.002126	-0.006375	-0.001564	0.000891	0.006486
30Y	-0.000302	0.002159	-0.006604	-0.001533	0.000854	0.006174

Table 44: Out-of-sample reconstruction residuals — Hybrid, $N = 853$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000162	0.002465	-0.010921	-0.001130	0.001169	0.006747
2Y	-0.000141	0.002529	-0.010405	-0.001287	0.001122	0.007330
3Y	-0.000140	0.002565	-0.009992	-0.001402	0.001124	0.008060
4Y	-0.000151	0.002560	-0.009611	-0.001455	0.001137	0.008315
5Y	-0.000165	0.002523	-0.009260	-0.001493	0.001122	0.008272
6Y	-0.000179	0.002482	-0.008908	-0.001461	0.001119	0.008346
7Y	-0.000193	0.002442	-0.008714	-0.001446	0.001057	0.008300
8Y	-0.000205	0.002414	-0.008537	-0.001432	0.001039	0.008296
9Y	-0.000215	0.002391	-0.008403	-0.001391	0.001020	0.008310
10Y	-0.000221	0.002365	-0.008268	-0.001419	0.000982	0.008322
12Y	-0.000240	0.002345	-0.008005	-0.001396	0.000935	0.008384
15Y	-0.000262	0.002290	-0.007564	-0.001465	0.000881	0.007700
20Y	-0.000287	0.002257	-0.006728	-0.001497	0.000855	0.008093
25Y	-0.000320	0.002171	-0.006069	-0.001584	0.000853	0.006943
30Y	-0.000324	0.002195	-0.006347	-0.001570	0.000816	0.006772

Table 45: Out-of-sample reconstruction residuals — Hybrid random walk, $N = 853$.

Tenor	Mean	Std. Dev.	Min	Q25	Q75	Max
1Y	-0.000179	0.002460	-0.010834	-0.001091	0.001149	0.006841
2Y	-0.000170	0.002517	-0.010308	-0.001263	0.001051	0.007507
3Y	-0.000176	0.002552	-0.009891	-0.001416	0.001112	0.007457
4Y	-0.000190	0.002548	-0.009510	-0.001480	0.001127	0.007558
5Y	-0.000206	0.002513	-0.009162	-0.001500	0.001133	0.007672
6Y	-0.000220	0.002474	-0.008887	-0.001487	0.001120	0.007739
7Y	-0.000234	0.002436	-0.008708	-0.001485	0.001068	0.007696
8Y	-0.000245	0.002410	-0.008535	-0.001458	0.001047	0.007692
9Y	-0.000254	0.002388	-0.008405	-0.001489	0.001031	0.007700
10Y	-0.000259	0.002362	-0.008263	-0.001531	0.000990	0.007698
12Y	-0.000278	0.002339	-0.008002	-0.001506	0.000904	0.007845
15Y	-0.000299	0.002288	-0.007638	-0.001519	0.000875	0.007082
20Y	-0.000323	0.002248	-0.006823	-0.001528	0.000849	0.007459
25Y	-0.000353	0.002163	-0.006375	-0.001618	0.000863	0.006486
30Y	-0.000357	0.002190	-0.006604	-0.001604	0.000843	0.006174

References

- Ayitey Junior, M., Appiahene, P., and Appiah, O. (2022). Forex market forecasting with two-layer stacked long short-term memory neural network (lstm) and correlation analysis. *Journal of Electrical Systems and Information Technology*, 9(1):14.
- Bie, S., Diebold, F. X., He, J., and Li, J. (2024). Machine learning and the yield curve: tree-based macroeconomic regime switching. PIER Working Paper Archive 24-028, Penn Institute for Economic Research, Department of Economics, University of Pennsylvania.
- BIS, B. o. I. S. (2005). *Zero-Coupon Yield Curves: Technical Documentation*, volume 25.
- Campbell, E. (2008). The ethics of teaching as a moral profession. *Curriculum Inquiry*, 38(4):357–385.
- Cleveland, W. S. (1979). Robust locally weighted regression and smoothing scatterplots. *Journal of the American Statistical Association*, 74(368):829–836.
- Diebold, F. X. and Li, C. (2006). Forecasting the term structure of government bond yields. *Journal of Econometrics*, 130(2):337–364.
- Fischer, T. and Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2):654–669.
- Gers, F. (1999). Learning to forget: continual prediction with lstm. pages 850–855.
- Gilli, M., Grosse, S., and Schumann, E. (2010). Calibrating the nelson-siegel-svensson model. *SSRN Electronic Journal*.
- Goodfellow, I., Bengio, Y., and Courville, A. (2016). *Deep Learning*. MIT Press. <http://www.deeplearningbook.org>.
- Greff, K., Srivastava, R., Koutník, J., Steunebrink, B., and Schmidhuber, J. (2015). Lstm: A search space odyssey. *IEEE transactions on neural networks and learning systems*, 28.
- Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9:1735–1780.
- Kingma, D. P. and Ba, J. (2017). Adam: A method for stochastic optimization.
- Litterman, R. and Scheinkman, J. (1991). Common factors affecting bond returns. *Journal of Fixed Income*, 1(1):54–61.
- Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer.
- Nelson, C. R. and Siegel, A. F. (1987). Parsimonious modeling of yield curves. *The Journal of business (Chicago, Ill.)*, 60(4):473–489.
- Nunes, M., Gerding, E., McGroarty, F., Niranjana, M., and Sermpinis, G. (2026). Deep learning for bond yield forecasting: The lstm-laglasso. *International Journal of Finance and Economics*, 31(1):1269 – 1283. Cited by: 1; All Open Access, Green Open Access, Hybrid Gold Open Access.

- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., and Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(56):1929–1958.
- Suimon, Y., Sakaji, H., Izumi, K., and Matsushima, H. (2020). Autoencoder-based three-factor model for the yield curve of japanese government bonds and a trading strategy. *Journal of Risk and Financial Management*, 13(4).
- Svensson, L. E. (1994). Estimating and interpreting forward interest rates: Sweden 1992 - 1994. (4871).
- Verner, R., Tkáč, M., and Tkáč, M. (2021). Improving quality of long-term bond price prediction using artificial neural networks. *Kvalita inovácia prosperita*, 25(1):103–123.